Journal of Urban Economics 88 (2015) 50–66

Contents lists available at ScienceDirect

Journal of Urban Economics

www.elsevier.com/locate/jue

The local impact of typhoons on economic activity in China: A view from outer space

Robert J.R. Elliott [a,*], Eric Strobl [b], Puyang Sun [c]

[a] University of Birmingham, UK
[b] Ecole Polytechnique & IPAG, France
[c] Nankai University, China

ARTICLE INFO

Article history:
Received 7 October 2014
Revised 20 March 2015
Available online 5 June 2015

JEL classification:
O17
O44
Q54

Keywords:
China
Typhoons
Wind field model
Economic impact
Nightlight imagery

ABSTRACT

We examine the impact of typhoons on local economic activity in coastal China. To capture potential damages from an individual typhoon we use historical typhoon track data in conjunction with a detailed wind-field model. We then combine our damage proxy with satellite derived nightlight intensity data to construct a panel data set that allows us to estimate the impact of typhoons at a spatially highly disaggregated level (approx. 1 km). Our results show that typhoons have a negative and significant, but short-term, impact on local activity – a typhoon that is estimated to destroy 50% of the property reduces local economic activity by 20% for that year. Over our period of analysis (1992–2010) total net economic losses are estimated to be in the region of $US 28.34 billion. To assess the damage risk from future typhoons we use simulated probability distributions of typhoon occurrence and intensity and combine these with our estimated effects. Results suggest that expected annual losses are likely to be around $US 0.54 billion.

1. Introduction

Annual global damages arising from tropical cyclones have been estimated to be around $US 26 billion.[1] In this regard China, with on average seven typhoons per year making landfall resulting in annual damages of about $US 5.6 billion according to official figures, has been a major contributor to the global damage estimates.[2] For example, in 2006 China was struck by Typhoon Soamai which subjected the provinces of Zhejiang and Fujian to winds of up to 135 mph and an associated storm surge of up to 10 feet. As a consequence, Typhoon Soamai destroyed thousands of buildings, sunk over 1000 boats, led to power cuts in six cities, and displaced up to 1.7 million people. Total costs were estimated by the Chinese government to be in the region of $US 1.4 billion. Such potentially high losses and the subsequent interruption to economic activity raises the question of what impact typhoons ultimately have had on the Chinese economy and what the future impact maybe.[3] The contribution of this paper is to provide the first detailed study of the economic effects of typhoons for China.[4]

In seeking guidance on how to conduct an analysis for China it is noteworthy that while the literature examining the economic impact of natural disasters is still relatively new, there has been relatively rapid evolution in both the approach and methodology used. The majority of the earlier studies conduct global cross-country analyses of the impact of natural disasters in general and find mixed results; see the reviews by Cavallo and Noy (2011)

* Corresponding author at: Department of Economics, University of Birmingham, UK.

E-mail address: r.j.elliott@bham.ac.uk (R.J.R. Elliott).

[1] See Mendelsohn et al. (2012). Neumayer and Barthel (2011) point out that over the last 30 years annual global economic losses due to weather-related events have increased at a rate of $US 2.7 billion a year with direct physical damage from natural disasters estimated to be $US 1527.6 billion between 1975 and 2008 (UNISDR, 2009).

[2] See Liu et al. (2001) for a remarkable 1000 year history of typhoon strikes in Guangdong Province and Louie and Liu (2003) for a discussion of the earliest records of typhoon activity in China. Other China specific studies that concentrate on the physical impact of typhoons include Gao et al. (1999), Chen et al. (2009), Meng et al. (2007), Wang et al. (2007), Zhang et al. (2009a,b) and Cao and Wang (2013).

[3] Typhoons are defined as tropical cyclones of the western North Pacific (NWP) that maintain a maximum wind speed for one minute of 33 ms^{-1} or over.

[4] There are two studies that examine the economic impact of natural disasters in general in China, but without specifically identifying the impact of tropical cyclones. Vu and Noy (2013) find that natural disasters, as derived from EM-DAT (Emergency-Events Database), affect both provincial income and investment, although their effect depends on the measure used. Similarly, the results in a province level study of the impact of disasters on output by Vu and Hammes (2010) depends heavily on the measure of natural disaster used.

http://dx.doi.org/10.1016/j.jue.2015.05.001
0094-1190/

and Felbermayr and Gröschl (2013). A small number of studies focus specifically on tropical storms. For example, Strobl (2012) shows that tropical storms in the Central American and Caribbean region cause country growth rates to fall by 0.6 percentage points in the year of the strike. It should also be noted that many of the earlier studies, including Vu and Hammes (2010) and Vu and Noy (2013) for China, tend to use *ex-post* collected data on damages, deaths, and/or the affected area to capture the impact of natural disasters. However, Strobl (2012) shows that these data are likely to introduce measurement and omitted variable bias into the estimation and recommends the use of actual physical characteristics, such as wind speed for tropical storms, to proxy potential damages.

Another important development in the literature has been the recognition that natural disasters are inherently local phenomena, so that their overall impact may be fully or partially 'aggregated out' at the cross-country or large regional level, particularly for large countries. Bouwer (2011) notes when talking about the shortcomings of the existing literature that "*...scale of analysis is also an issue, because aggregating to the regional or global level may have the advantage that local variability is eliminated, but one could fail to see trends [...] that may vary per location in sign and magnitude*" (pg. 43). This was empirically shown for the US by Strobl (2011) who found hurricanes to have an annual negative impact at the county level, but no effect at the state or national level.[5]

In this paper we set out to identify the economic effect of tropical storms on coastal China by taking into account and refining recent developments in the literature. More specifically, we first combine actual typhoon track data with a wind field model to derive an *ex-ante* index of potential damage due to typhoons at any point relative to the storm. We then, in order to address the potential spatial aggregation bias associated with using national or large regional regions as the unit analysis, use time-varying estimates of night-light intensity from satellite imagery as proxies of economic activity (Chen and Nordhaus, 2011; Henderson et al., 2011). Note that there are now a number of papers that specifically use nightlight imagery to proxy local economic wealth; see, for instance, Harari and La Ferrara (2013), Hodler and Raschky (2014) and Michalopoulos and Papaioannou (2014). As a matter of fact, a recent study of the Caribbean by Bertinelli and Strobl (2013) demonstrated that the level of spatial aggregation of nightlight imagery can more accurately capture the effect of hurricane damage. An additional contribution for China is that we also allow for a different impact of typhoons on nightlights in storm surge prone areas, an aspect that has been previously ignored in the literature. Combining our income proxy with local potential damage measures thus provides us with a large annual panel of data covering nearly twenty years with which we can estimate the impact of typhoons on local net economic activity in China.

Although the literature on the economic impact of tropical storms, and natural disasters in general, has improved in terms of methodology and data in recent years, one shortcoming has been that the implications of the results for risk analysis is still relatively limited. More specifically, tropical storms, like most natural disasters, particularly the extreme ones, tend to be relatively infrequent events. Thus, while one may obtain an accurate estimate of the economic impact of tropical cyclones with available historical data, these data do not provide enough observations to derive reliable probability distributions of tropical storms' intensity and occurrence. Traditionally this problem has been addressed by using extreme value theory or other statistical methods. However, since the historical data typically contain only a few very strong tropical storm strikes the estimated probabilities can be very sensitive to the tail of the assumed distribution, making direct inference fairly unreliable, particularly for local areas. Here we follow Emanuel et al. (2008) and instead rely on a large set of synthetic tracks generated from a random draw using a space time probability density function of tropical cyclone formation based on a coupled ocean–atmosphere tropical storm model. This allows us to derive a statistically sound distribution of hypothetical hurricane storms and accompanying wind strengths near the Chinese coast according to recent climatological features. We then combine these synthetic tracks with our econometric estimates of a typhoon's local impact to provide local typhoon damage risk measures for coastal China.

Arguably China is a particularly attractive case study. First, growth in exposure to typhoons as a result of rapid economic development and migration patterns and a growing insurance market in China has increased the interest from policymakers and the insurance industry to better understand the risks associated with typhoons (Freeman et al., 2003; Ward et al., 2008; Neumayer and Barthel, 2011).[6] According to ADRC (2013), over 70% of China's cities and 50% of the population are located in regions that frequently experience major natural disasters. Second, although the science is currently inconclusive, the possibility that global warming will result in more numerous or more intense typhoons in the future increases the need for policymakers to gain a better understanding of the current relationship between typhoons and economic costs (Webster et al., 2005; Landsea et al., 2006; Elsner et al., 2008; Li et al., 2010). Finally, the net economic impact caused by any natural disaster is closely linked to a country's disaster preparedness (Kahn, 2005). In China, disaster management falls under the China National Committee for Disaster Reduction (NCDR), which is comprised of 34 ministries and departments and includes the relevant military agencies and social groups and its main function is to coordinate across agencies and to instigate plans and policies for disaster mitigation.[7] Importantly, disaster management strategies in China are often considered to be remarkably successful, which may have dampened any negative impact on economic activity due to typhoons.[8,9]

[5] As typhoons are part of the atmospheric and hydrological process typhoons are also likely to reoccur and hence have a cumulative effect on development (Benson et al., 2000).

[6] Zhang and Zhu (2005) discuss the development of the insurance industry in China. According to Swiss Reinsurance Company (2005) China's premium income was $US 52.17 billion (3.26% of GDP) in 2004 up from $US 1.37 billion (0.7% of GDP) in 1986 (Swiss Reinsurance Company, 2005). The number of firms in the property insurance industry has also increased from 8 in 1996 to 26 in 2004 (Wu, 2004). China is currently testing a new insurance system to fund reconstruction work after natural disasters which will share the cost between the government, insurance and reinsurance firms and individuals.

[7] The first national disaster reduction plan was written in conjunction with the ninth five-year plan and a state emergency response system built and developed as part of the twelfth five-year plan (2011–2015). The priority areas are disaster prevention and relief management which includes scientific research, technological advances and education. Coordination across local and central government is also an important element of disaster management (ADRC, 2013). Satellites for remote sensing have also recently been launched for disaster monitoring and assessment.

[8] The Ministry of Civil Affairs is currently responsible for disaster relief. The contingency plan for disaster relief includes plans issued at the province, city and county-level, as well as for individual towns, schools and factories. In addition stockpiles for disaster relief have been built up across 22 cities and smaller disaster prone localities in China. A campaign of public awareness has also been undertaken and a booklet "Handbook of Disaster Prevention and Self-Rescue" has been published. There are also plans for a well-equipped national chain of emergency shelters that will include schools and stadiums and other public buildings. Moreover, rural and urban communities will have their own emergency response plan (ADRC, 2013). Following the Sichuan (2008) earthquake the Chinese government obtained resources from across the country and a three-year target was announced that all homeless households would be rebuilt. Grant were also provided to households that were homeless and could also apply for additional loans. The subsidy and loan system was directed by the central government but the implementation was at the discretion of county governments and village committees (Tse et al., 2013).

[9] One example is the adaptation of capital (housing and infrastructure) in disaster prone areas which has led to casualties from China's frequent floods falling dramatically from over 140,000 and 33,000 lives in 1931 and 1954 respectively to 3000 for a similar flood in 1998 (Annan, 1999).

To briefly highlight the results of our paper we find that over the time period 1992–2010 typhoons had a significant and negative impact in China. The headline result is that a typhoon that is estimated to damage to property in the region of 50% will result in a 20% reduction in local economic activity for that year. Our estimate of the total net economic loss for our sample period is around $US 28 billion (or $US 1 billion annually), with the value for 2005 alone being over $US 5.63 billion. Rough estimates show that about 36% of this damage estimate was due to increased coastal exposure since 1992 as a result of China's rapid economic growth during this period. Our risk analysis based on future synthetic tracks for typhoons, under the assumption of present day weather conditions, suggests that expected annual losses are likely to be around $US 0.54 billion in coastal China, but may differ widely along the coastline.

The remainder of this paper is organized as follows. Section 2 describes our data and methodology, while Section 3 presents our econometric results. In Section 4 we translate our estimates into monetary loss figures. Section 5 contains our risk analysis. Finally Section 6 concludes.

2. Data and summary statistics

2.1. Geographic region

Given that tropical storms lose speed relatively quickly after they make landfall we focus on the economic impact on China's coastal area of the West Pacific, i.e. the region most likely to be affected by typhoon strikes.[10] Moreover, since part of our goal is to translate the impact of typhoons on local economic activity as measured by nightlights into monetary values and the available regional data on actual income per capita is at the Chinese "City" level, we restrict our analysis to coastal cities. We define a coastal county-city to be a city whose centroid is within 50 km of the shoreline.[11] One should note that Chinese city level regions are administrative areas more akin to a county in the US or the UK and does not refer to urbanized areas only.

Of the 340 Chinese county-cities in our dataset 60 are defined as coastal and are depicted by the darker spatial areas in Fig. 1. The major coastal cities include Shanghai, Tianjin, Guangzhou, Dalian City (Liaoning Province), Qingdao (Shandong Province), Ningbo City (Zhejiang Province) and Fuzhou City (Fujian Province).[12] In comparing coastal to non-coastal city-counties, the summary statistics in Table 1 show that the former are on average smaller in geographical size, but not surprisingly have higher population densities. In terms of income, coastal cities are substantially richer, measured both by GDP per square kilometer or in GDP per capita. In relative terms China's coastal cities' constitute approximately 10 per cent of China's physical area, 22 per cent of its population, but a considerable 42 per cent of its total income. Finally, one may want to note, that although 'cities' in China are administrative units like counties in the US, they are much larger. For instance, the average size of one of the 3007 US counties is approximately 1610 km^2, with a population density of 38 people per km^2 compared to 8050 km^2 and 663 people per km^2 for area and population density for the average Chinese coastal county-city respectively.

2.2. Economic activity

2.2.1. Nightlights

In order to proxy economic activity at the local level we use data derived from satellite images of nightlights. More specifically, we use nightlight imagery provided by the Defense Meteorological Satellite Program (DMSP) satellites. In terms of coverage each DMSP satellite has a 101 min near-polar orbit at an altitude of about 800 km above the surface of the earth, providing global coverage twice per day, at the same local time each day. In the late 1990s, the National Oceanic and Atmospheric Administration (NOAA) developed a methodology to generate "*stable, cloud-free nightlight data sets by filtering out from the data transient light produced by for example forest fires and other random noise events occurring in the same place less than three times*" (see Elvidge et al., 1997 for a detailed description of the filtering process). The resulting images provide the percentage of nightlight occurrences for each pixel per year normalized across satellites to a scale ranging from 0 (no light) to 63 (maximum light). The spatial resolution of the original images was about 0.008 degrees on a cylindrical projection (i.e., with constant areas across latitudes) and was later converted to a polyconic projection, giving squares of about 1 km^2 near the equator. In order to get yearly values, simple averages across daily (filtered) values of grids were generated. Data are publicly available on an annual basis for the period 1992–2010.[13]

The annual nightlight data from these satellites essentially captures nocturnal human activity, such as electrified human settlements and gas flares, which we take as a proxy for local economic activity. Fig. 2 present images of nightlights for 1992 and 2010, where the outline of our coastal county-cities are depicted in red. The brightly lit areas in Figs. 2 and 3 correspond to centers of economic activity, where, for instance, the cities of Hong Kong, Beijing, Chongqing and Shanghai stand out. Noteworthy is the dramatic increase in nightlight intensity over time, a trend that is particularly pronounced for coastal areas.

One consequence of using cells as indicators of economic activity is that, as Fig. 2 illustrates, a significant proportion of 1 km^2 cells are unlit, i.e. have a value of zero for our entire sample period. We assume that these cells do not capture any economic activity and thus we exclude them from our analysis. For all other cells, i.e. those with at least one non-zero value for the sample period, we keep all non-zero observations in addition to those zero values that immediately follow a positive value. The latter is to allow for the possibility that typhoon damage may have reduced economic activity to the extent that nightlights were no longer observable by the satellites. Since we do not know whether zero values in 1992 follow positive values in 1991 we exclude 1992 from our analysis. This leaves us with a sample of 563,192 cells with an average of 16 observations for each cell. Table 2 summarizes our nightlight data. As can be seen, the average nightlights value is 10.63 albeit with a relatively large standard deviation.

2.2.2. City GDP data

We supplement our analysis with county-city level GDP figures from the National Bureau of Statistics (2013) for the period 1993–2008. Fig. 4 presents a scatter plot of the GDP/km^2 and average

[10] In a study of the US, Strobl (2011) shows that tropical storms have a negative effect on coastal counties but no impact on counties that lie further inland. US counties are also smaller on average than Chinese-county cities.

[11] There is no consensus on the definition of a coastal city. For example, Polyzos and Tsiotas (2012) suggest a 40 km wide zone from the coastline in a study of the dynamics of coastal cities in Greece. Nicholls (2006) defines a "near-coastal zone" as being both within 100 m elevation and within 100 km of the coast.

[12] In 2007 China had 656 recorded urban cities (slightly below the 2006 number as five cities were absorbed into nearby larger cities (Urban Development Report of China) 45 of which had a population in 2007 of above one million people and 286 cities with a population below 200,000. By 2025 it is predicted that China will have 221 urban cities with populations greater than one million people (compared to 35 in Europe today) (McKinsey, 2009).

[13] For the years when satellites were replaced observations were available from both the new and old satellite. In this paper we use the imagery from the most recent satellite but as part of our sensitivity analysis we also re-estimated our results using an average of the two satellites and the older satellite only. The results of these latter two options were almost quantitatively and qualitatively identical.

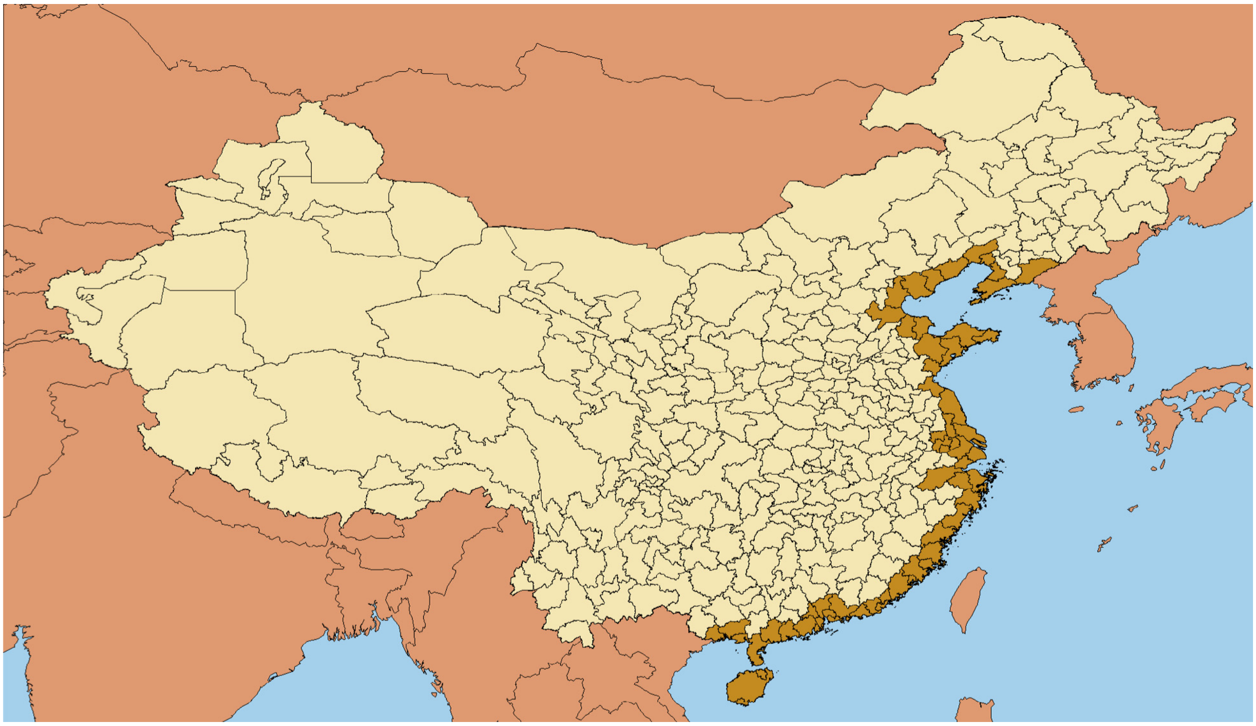

Fig. 1. Coastal cities of China. Notes: (1) Chinese 'cities; (2) coastal cities in beige. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Summary statistics of coastal and non-coastal cities.

	Coastal		Non-coastal	
	Mean	St. Dev.	Mean	St. Dev.
AREA	8050	4374	18,551	24,138
POP/AREA	663	629	362	249
GDP/POP	29	38	13	11
GDP/AREA	207	365	47	60
INVESTMENT/AREA	405	760	214	387

Notes: (a) Area is in square kilometers, (b) GDP is measured in real 2011 thousands of RMB.

Nightlight intensity/km^2 values at the coastal county-city level for 4 years which span our period of analysis. The immediate observation is that there appears to be a strong positive correlation between these two variables for each year and provides additional support for our use of nightlights as a proxy for local income. All nominal values are deflated to $US in 2013 prices.

2.3. Wind field modeling

2.3.1. Typhoon track data

Our source for typhoon track data is the Regional Specialized Meteorological Centre (RSMC) Best Track Data, which since 1951 has provided six hourly data on all tropical cyclones in the West Pacific with maximum sustained wind speed of at least 119 km/h. Other information includes the position of the eye of the storm, central pressure and the maximum wind speed. We linearly interpolate these to three hourly positions. We also restrict the set of storms to those that came within 500 km of the coast of China (since tropical cyclones generally do not exceed a diameter of a 1000 km) and to those that at some stage achieved at least typhoon strength. Fig. 5 presents the tracks of all remaining tropical storms for our sample period 1992–2010, where the red portion of the tracks refers to the segment of the storm that reached typhoon strength. As can be seen, it is mostly the southern coastal areas that are affected by typhoons. All in all a total of 110 typhoon strength storms traversed the area during our time period. The strongest, measured in terms of maximum wind speed, was the aforementioned Typhoon Saomai in 2006, which reached maximum winds of up to 216 km/h.

2.3.2. Wind field model

What level of wind a particular location will experience during a passing typhoon depends crucially on that location's position relative to the storm and the storm's movement and features. Thus, to properly assess local damage due to typhoons requires explicit wind field modeling. In order to calculate the wind speed experienced due to typhoons we use Boose et al.'s (2004) version of the well-known Holland (1980) wind field model. More specifically, the wind experienced at time t due to typhoon j at any point $P = i$, i.e., V_{ijt}, is given by:

$$V_{ijt} = GF\left[V_{m,jt} - S(1-\sin(T_{ijt}))\frac{V_{h,jt}}{2}\right]\left[\left(\frac{R_{m,j,t}}{R_{it}}\right)^{B_{jt}} \exp\left(1-\left[\frac{R_{m,j,t}}{R_{it}}\right]^{B_{jt}}\right)\right]^{\frac{1}{2}} \tag{1}$$

where V_m is the maximum sustained wind velocity anywhere in the typhoon. T is the clockwise angle between the forward path of the typhoon and a radial line from the typhoon center to the point of interest, $P = i$, V_h is the forward velocity of the tropical storm, R_m is the radius of maximum winds, R is the radial distance from the center of the tropical storm to point P. The relationship between these parameters and point $P = i$ are depicted in Fig. 6. The remaining ingredients consist of the gust factor G and the scaling parameters F, S, and B, for surface friction, asymmetry due to the forward motion of the storm, and the shape of the wind profile curve, respectively.

In terms of implementing (1) note that V_m is given by the storm track data, V_h can be directly calculated by following the storm's movements between locations, and R and T are calculated relative

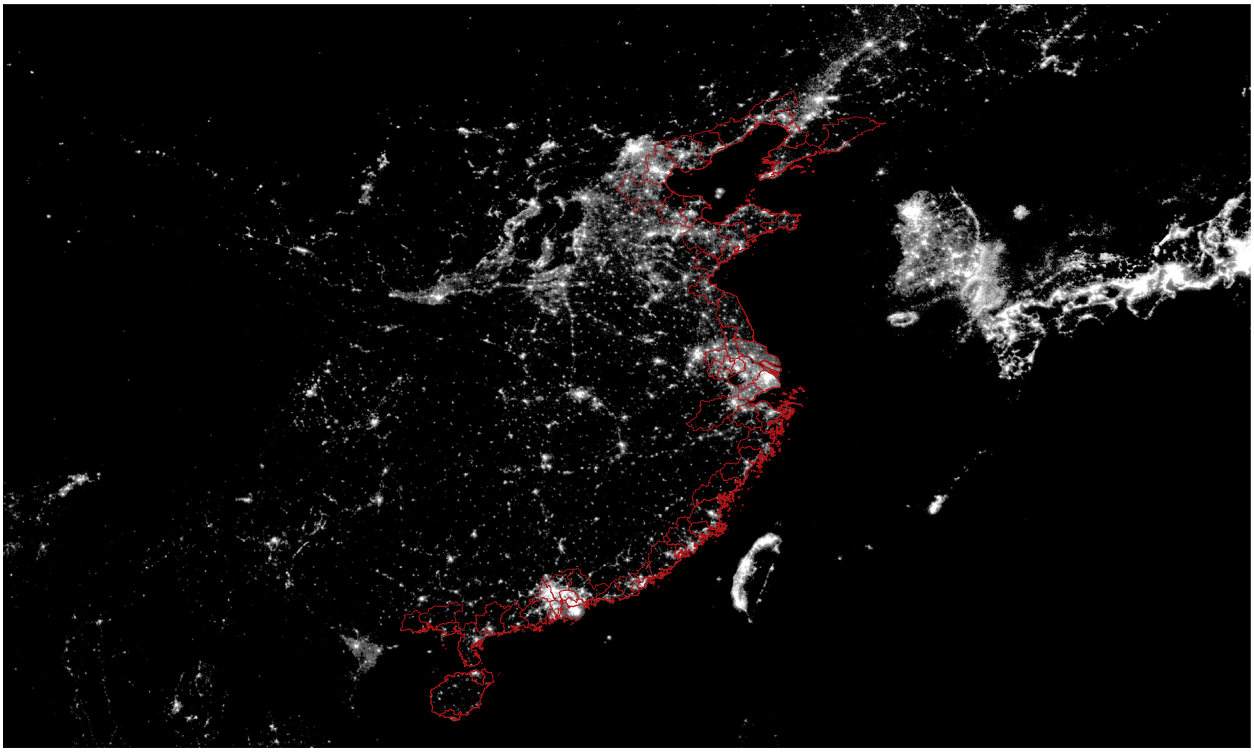

Fig. 2. Nightlight imagery from the DMSP satellites in 1992. Notes: (1) Nightlight intensity in 1992; (2) coastal cities outline in red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

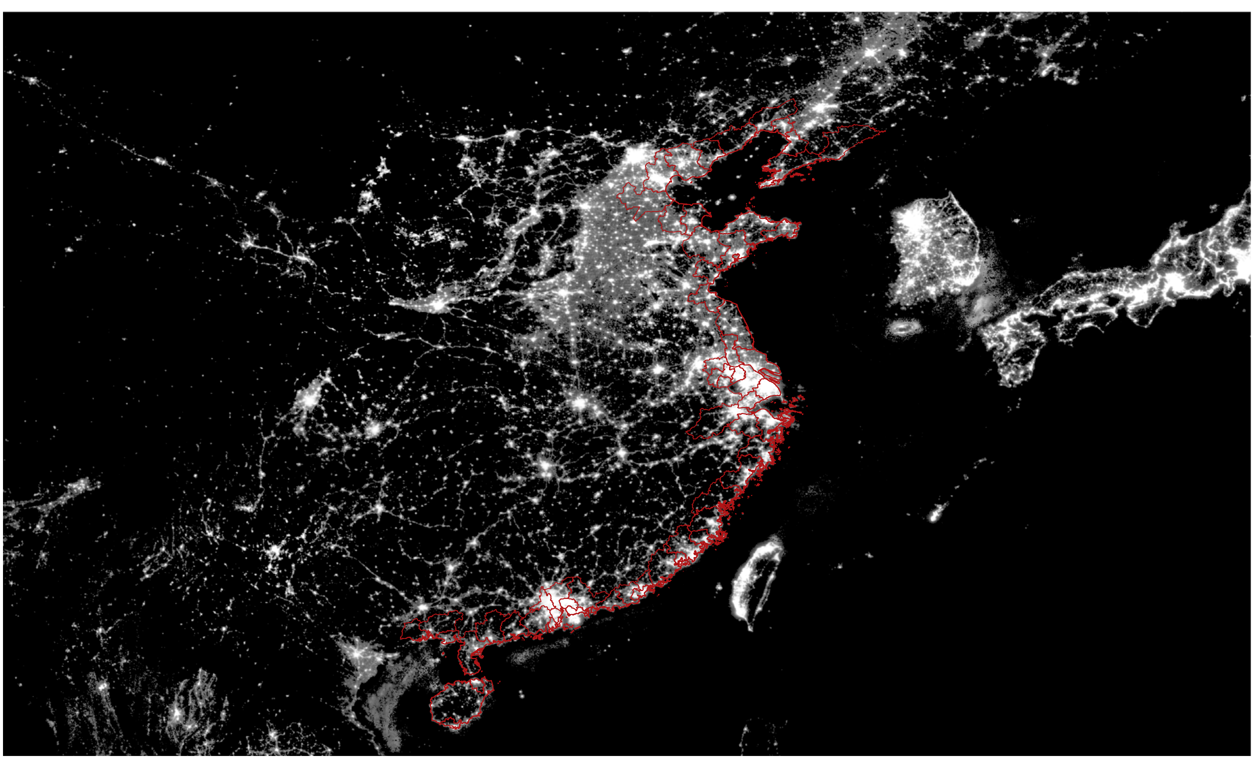

Fig. 3. Nightlight imagery from the DSMP satellites in 2010. Notes: (1) Nightlight intensity in 1992; (2) coastal cities outline in red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

to the point of interest $P = i$. All other parameters have to be estimated or assumed. For instance, we have no information on the gust wind factor G. However, a number of studies (e.g. Paulsen and Schroeder, 2005) have measured it explicitly and suggest that it is around 1.5, and hence, we similarly assume it to be 1.5. In addition, we have no information on the surface friction to directly

Table 2
Summary statistics for the satellite imagery nightlight data.

Variable	Mean	Minimum	Maximum	Standard deviation
Nightlights	10.63	0	63	13.81
f (≠0)	0.019	2.41e−20	0.639	0.037
f (≠0) in storm surge areas	0.021	7.01e−20	0.639	0.042
f (≠0) in non-storm surge areas	0.016	2.51e−17	0.198	0.024

Notes: (1) f index calculated as in Eq. (7). (2) f (≠0) are all non-zero values of f.

determine F. However, in the studies cited in their review of tropical storm hazard modeling, Vickery et al. (2009) suggest that in open water the reduction factor was about 0.7 which means a reduction of about 14% on the coast and 28% 50 km inland. We thus adopt a reduction factor that linearly decreases within this range as we consider points i further inland. Following Boose et al. (2004) we assume S to take on a value of 1. For the remaining parameters, Vickery and Wadhera (2008) emphasize that the radial pressure profile parameter B and the radius maximum winds R_{max} play an important role in estimating local wind speeds. Thus, we estimate these rather than using assumed values. More specifically, for B we employ Holland's (2008) approximation method:

$$B = b_s\left(\frac{v_{mg}}{v_m}\right)^2 \approx 1.5b_s \tag{2}$$

where

$$b_s = -4.4x10^{-5}\Delta p^2 + 0.01\Delta p + 0.03\frac{\partial p_c}{\partial t} - 0.014\psi + 0.15V_T^x + 1.0 \tag{3}$$

$$\text{and} \quad x = 0.6\left(1 - \frac{\Delta p}{215}\right) \tag{4}$$

where Δp is the pressure drop due to the cyclone center, $\delta p_c/\delta t$ is the intensity change, ψ is the absolute value of latitude, V_t is the cyclone transition speed, and v_{mg}/v_m is the conversion factor from gradient to surface wind.[14]

In order to derive a value for R_{max} we employ the parametric model estimated by Xiao et al. (2009) for Hong Kong:

$$\ln R_{max} = 5.3259 + -0.0249\Delta p - 0.0161\psi \tag{5}$$

where R_{max} is constrained to remain above 8 km and below 150 km.

Given the typhoon track data and the assumed parameters, outlined above, Eqs. (1)–(5) allows us to estimate the wind speed experienced at any point i, in our case the centroid of each 1 km nightlight cell, at each point in time for each typhoon.

2.4. Damage function

The damage due to tropical storms takes three main forms: (a) wind destruction, (b) flooding/excess rainfall, and (c) storm surge. Importantly these are all correlated with wind speed and hence wind speeds experienced can be used as a general proxy for potential damages due to tropical storms. To translate wind speed into potential damage, one should note that property damage due to tropical storm should vary with the cubic power of the wind speed experienced on physical grounds, and it is for this reason that previous studies have simply used the cubic power of wind speed as a destruction proxy.[15] However, there is likely to be a threshold below which there is unlikely to be any substantial physical damage (Emanuel, 2011). Moreover, the fraction of property damaged should approach unity at very high wind speeds. To capture these features we employ the index, f, proposed by Emanuel (2011) that proxies the fraction of property damaged:

$$f_{ijt} = \frac{v_{n,ijt}^3}{1 + v_{n,ijt}^3} \tag{6}$$

where

$$v_{n,ijt} = \frac{MAX[(V_{ijt} - V_{thresh}), 0]}{V_{half} - V_{thresh}} \tag{7}$$

where V_{ijt} is the wind experienced at point i at time t due to storm j as calculated in (1), V_{thresh} is the threshold below which no damage occurs, and V_{half} is the threshold at which half of the property is damaged. Following Emanuel (2011) we use a value of 93 km (i.e. 50kts) for V_{thresh} and a value of 278 km (i.e. 150kts) for V_{half}. We depict the damage profiles of f in Fig. 7 as well as three consecutive bars reflecting the wind associated with Saffir–Simpson Scales 1, 3, and 5.[16] Wind speeds classified as level 1 (119 km/h), 3 (178 km/h), and 5 (252 km/h) correspond to f values of about 0.003, 0.090, and 0.39 respectively.

With this framework and the local wind speeds constructed in the manner described above, we can now calculate out the potential damage due to typhoons experienced by each centroid over our sample period. We find that approximately 19% of our observations experienced at least some positive value of f. The summary statistics in Table 2 show that when the damage index took on a non-zero value, the mean value of f was relatively small, i.e. around 0.019. In other words, when there was potential damage due to typhoons this was, on average, around 1.9 per cent. Nevertheless, over our sample period, potential damage in certain location(s) reached as high as 64%. In Fig. 8 we also present the graphical distribution of the average non-zero value of f for each grid cell over our sample period. As can be seen, there is considerable heterogeneity in the damage experienced over geographical space. In particular the average damage experienced in the south and along the coast is much higher. If one looks at the city averages then those ranked at the top in terms of average damage over our sample period were Wenzhou, Maoming, and Ningende, while 16 or slightly more than one quarter of our coastal city-counties experienced no damage at all. Note that the undamaged cities contained about 40 per cent of all cells that were lit at least once during our sample period and 35% of total average annual economic activity as measured by nightlight intensity. Nevertheless, while these cities were not affected during our sample period, an examination of the historical tracks which are available as far back as 1950 show that there were several occasions in which typhoon strength storms made landfall or came close enough to have caused some degree of destruction. This suggests that the probability of typhoon damages is positive (non-zero) even for these cities and hence they should be included in our sample. Table 2 also includes the average damage values for areas susceptible to storm surge where we find higher average damages for nightlight intensity in those areas. We now explain what we mean by storm surge areas.

[14] Note that Holland (1980) uses a value of 1.6 for this conversion factor. Instead we use 1.5 in order to be consistent with the value we use for F in Eq. (1). Using 1.6 as an alternative made no noticeable qualitative or quantitative difference to our results.

[15] See Strobl (2011, 2012) for details.

[16] According to the NOAA scale 1 corresponds to "No real damage to building structures. Damage primarily to unanchored mobile homes, shrubbery, and trees. Also, some coastal road flooding and minor pier damage", scale 3 to "Some structural damage to small residences and utility buildings with a minor amount of curtain wall failures. Mobile homes are destroyed. Flooding near the coast destroys smaller structures with larger structures damaged by floating debris. Terrain continuously lower than 5 feet ASL may be flooded inland 8 miles or more.", and scale 5 to "Complete roof failure on many residences and industrial buildings. Some complete building failures with small utility buildings blown over or away. Major damage to lower floors of all structures located less than 15 feet ASL and within 500 yards of the shoreline."

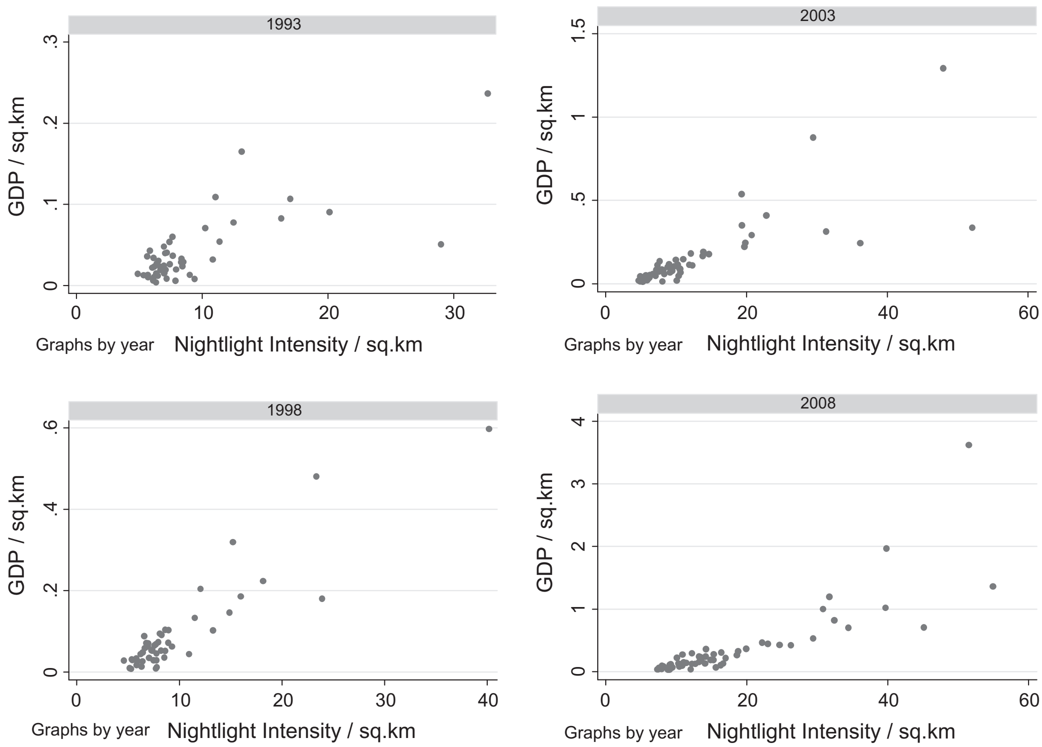

Fig. 4. GDP/km^2 versus nightlight intensity/km^2. Notes: (1) Plot of nightlight intensity per km^2 versus GDP per km^2 for cities in 1993, 1998, 2003, and 2008.

2.5. Storm surge prone areas

As noted earlier, one of the most damaging aspects of a typhoon can be storm surge. While the wind speed experienced is certainly correlated with storm surge, it is only imperfectly so. Unfortunately explicit storm surge modeling is inherently more complex than wind field modeling and requires geographical data that is not publicly available for China such as bathymetry, surface roughness, and information regarding the timing of tides. However, storm surge is only likely to be of relevance to an area

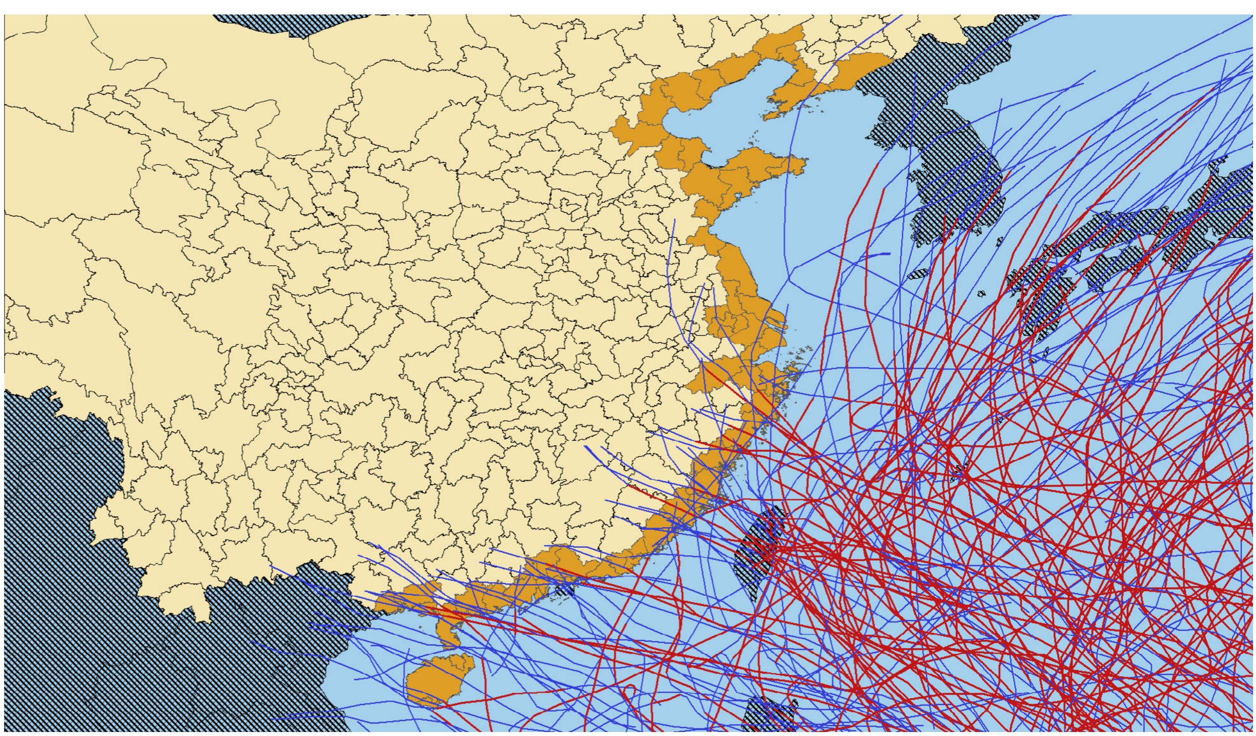

Fig. 5. Typhoon tracks of the coast of China 1992–2010. Notes: (1) Tracks of all tropical storms that reached at least typhoon strength during 1992–2010; (2) red portion indicates typhoons strength, blue portion indicates below typhoon strength. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

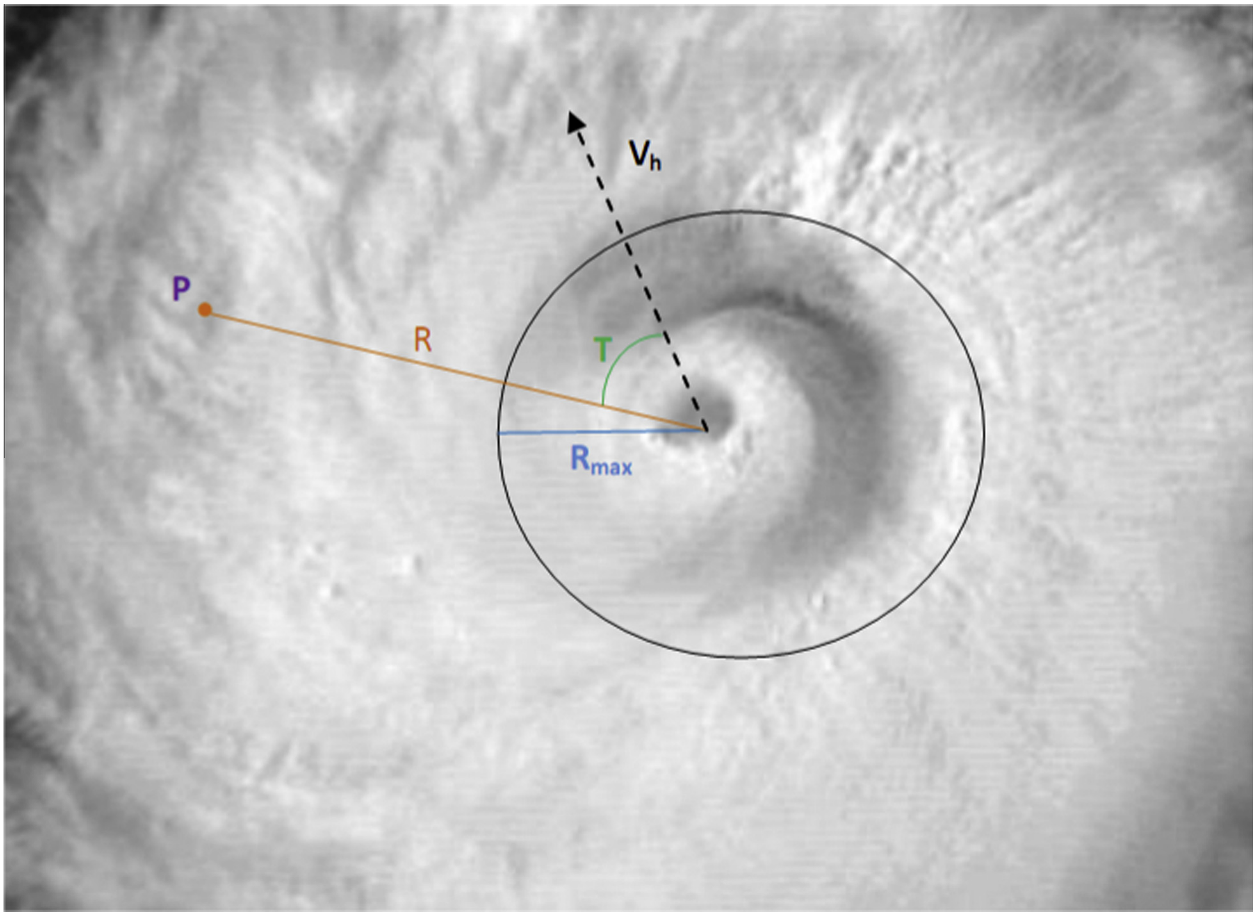

Fig. 6. Typhoon wind field model. Notes: (1) Sample diagram of input parameters into typhoon wind field model; (2) P: point of interest, R: distance from storm eye to point of interest, R_{max}: radius of maximum wind speed, T: angle of point relative to direction of storm; V_h: forward speed of storm.

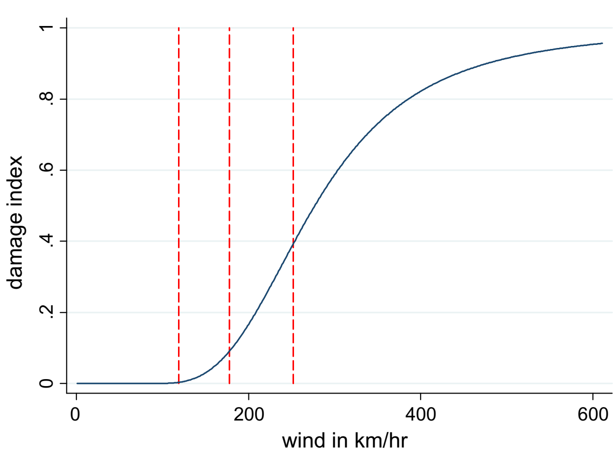

Fig. 7. Damage function (f). Notes: (1) Relationship between damage index (f) and wind speed; (2) red lines indicate (sequentially) thresholds for Saffir–Simpson Scale 1, 2, and 3. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

near to the shoreline with low enough contiguous elevation so that the area may be potentially submerged when there is a storm surge event. To identify low elevation coastal zones (LECZ) we follow McGranahan et al. (2007) and Brecht et al. (2012) and define land areas contiguous with the coastline up to a 10 m rise elevation using the Shuttle Radar Topography Mission (SRTM) elevation data set. Using this definition some LECZ areas are as far as 105 km inland although it is unlikely that storm surge would travel that far as its height would diminish as it moved inland. To determine the maximum distance inland a storm surge may travel we note that historically the maximum height of a storm surge ever recorded is believed to have been between 13 and 15 m.[17] If we take the latter value and assume a distance decay factor of 0.3 meters per kilometer (Brecht et al., 2012) we estimate that the maximum threshold inland distance is around 50 km.[18] Using this definition, about 24% of total Chinese coastal cities' area can be considered storm surge prone and are shown in Fig. 9. One observation is that the distribution of storm surge areas along China's coast is rather uneven.

3. Econometric analysis

3.1. Basic regression

The main purpose of this paper is to measure how the damage induced by typhoons affects local economic activity as measured by nightlight intensity. Our estimating equation is given by:

$$Nightlight_{it} = \alpha + \beta_f f_{it} + \pi_t + \mu_i + \varepsilon_{it} \tag{8}$$

[17] This was due to the Tropical Cyclone Mahina, a storm with sustained winds in excess of 280 km/h and struck Bathurst Bay, Australia on March 5, 1899.

[18] For example, evidence for the storm surge for Hurricane Katrina was found up to 20 km inland (Fritz et al., 2007) and close to 50 km for Hurricane Ike (Forbes et al., 2011).

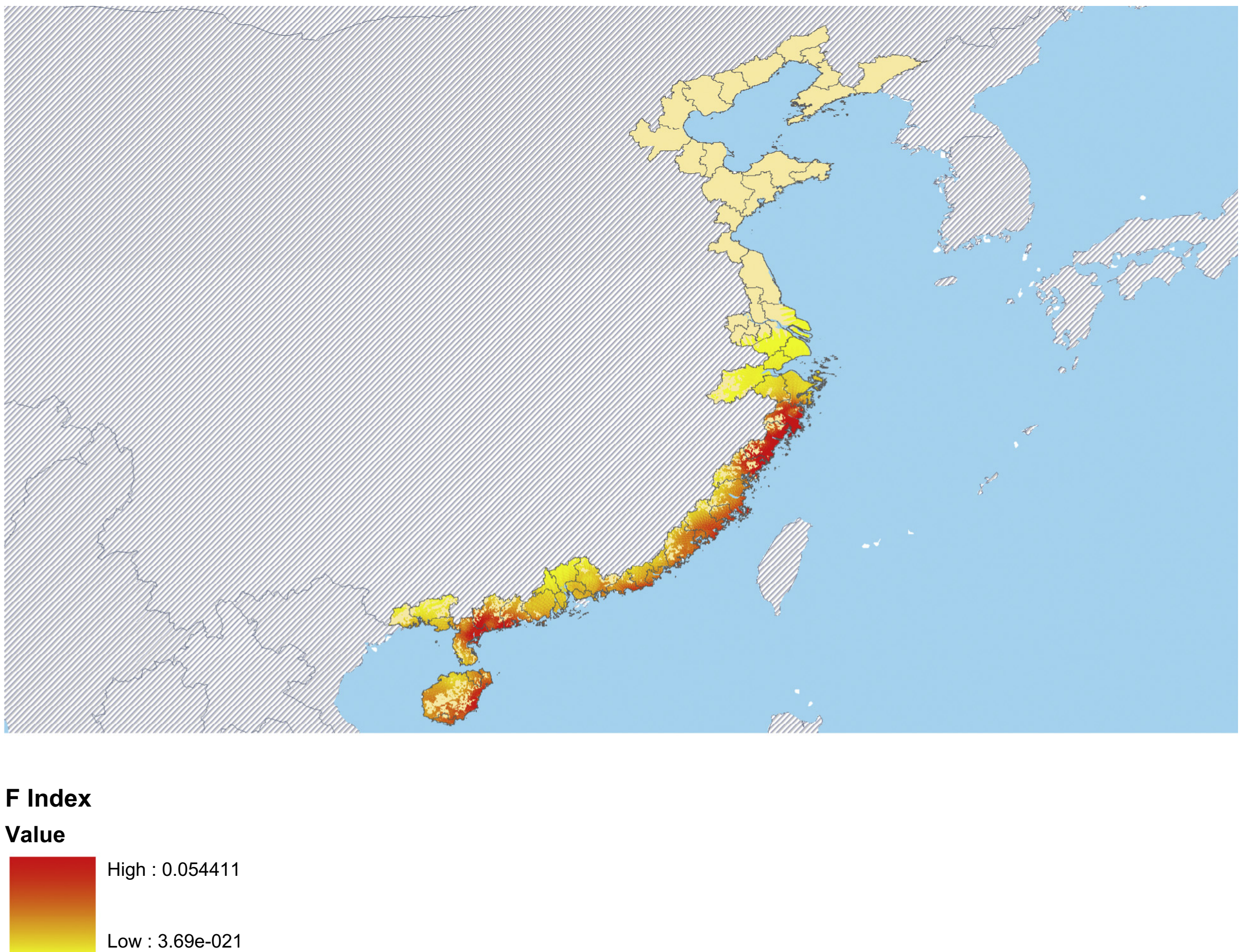

Fig. 8. Distribution of the average value of f index at the grid cell level 1992–2010.

where *Nightlights* is the nightlight intensity of cell i in year t, f is our damage index, and ε is the usual error term. The variable μ_i allows for unobserved cell time invariant effects that might be correlated with both typhoon destruction and economic activity. More precisely, whilst one can convincingly argue that the actual typhoon incidence can be considered to be an exogenous shock, it may be that some areas are more prone to typhoons than others and that economic agents know this and hence may invest more in damage prevention and/or are less likely to locate economic activity in those areas. We control for this possibility by using a fixed effects estimator. We also include time dummies π_t to account for time varying common changes in economic factors, such as aggregate disaster mitigation investment. These dummies also allow us to take into account changes in satellites and their productivity and accuracy as they age (previously discussed in footnote 13). To allow for correlation among economic outcomes across cells within county-cities we use robust standard errors clustered at the county-city level.[19] Finally, we employ standard panel unit root tests to investigate the possible non-stationarity in our nightlight data but found no evidence that this was a concern.[20]

We report the results from estimating Eq. (8) for our total coastal sample of cells in the first column of Table 3 where we assume V_{half} = 278 km/h. Observe that the coefficient on the typhoon damage index f is negative and significant. The size of the coefficient suggests that if half of the property in a given cell area were destroyed (f = 0.5) then local nightlight intensity would decrease by approximately 2 units. In other words, if we take an average lit cell (with a value of nightlight intensity of around 10) and assume that nightlight intensity captures economic activity over a year, then the effect of a typhoon translates into about a 20 per cent reduction. In contrast, if we observed the maximum loss of 0.63 it would translate as roughly equivalent to a 30 per cent reduction in net annual economic activity. In column (2) we investigate whether there is a longer term impact of typhoons by including a lagged value of our wind field damage index but find no such effect.[21] One should note that this is a result that echoes the general finding in the literature of a limited long term effect of tropical storms on economic activity.[22]

Referring back to the summary statistics in Table 2 recall that in terms of the potential damage the average was about 30% higher in storm surge prone relative to non-storm surge prone areas. To see whether the former are also more sensitive to tropical storms we split our sample into these two cell categories and re-ran Eq. (7) separately with and without the lag of our damage index with the results shown in columns (3) through (6). Accordingly, in column (3) the resultant coefficient on f is about 26% higher than

[19] We also experimented with clustering standard errors at the zip code and the provincial level but this did not change our results qualitatively.

[20] We employed the Im et al. (2003) unit root test for those cells for which we had enough subsequent observations over time. Our test statistic −7.892 suggested that we can treat our data as stationary.

[21] We also experimented with additional lags but these also returned insignificant coefficients.

[22] The short term nature of the costs was first documented by Strobl (2008).

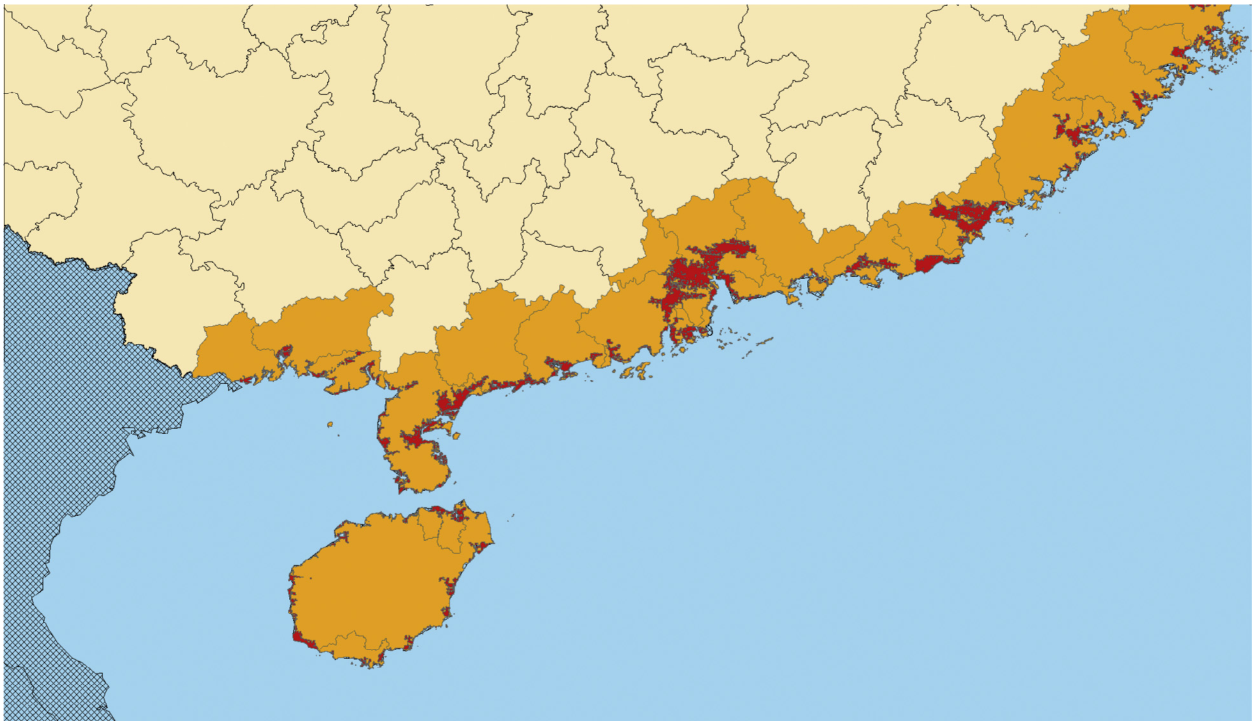

Fig. 9. Storm surge prone areas (concentrating on Southern China). Notes: (1) Red areas indicate storm surge prone areas. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Regression results (dependent variable nightlights) 1992–2010 (V_{half} = 278 km).

	(1)	(2)	(3)	(4)	(5)	(6)
f	−4.694**	−4.655**	−5.853*	−3.678*	−5.795*	−3.675*
	(1.729)	(1.741)	(2.725)	(1.576)	(2.643)	(1.586)
f(*t* − 1)		−2.399			−0.968	−2.382
		(2.211)			(2.666)	(1.841)
Sample	All	All	SSP	NSSP	SSP	NSSP
Obs.	9,173,598	9,173,598	2,425,384	6,748,214	2,425,384	6,748,214
Cells	563,192	563,192	140,212	422,980	140,212	422,980
R^2	0.310	0.310	0.397	0.283	0.397	0.283
F-test(b)	101.3	109.0	89.66	121.6	85.65	125.6

Notes: (1) Robust standard errors clustered at the city level in parentheses; (2) $^{**}p < 0.01$, $^{*}p < 0.05$; (3) time dummies included but not reported; (4) SSP: Storm Surge Prone Areas; NSSP: Non Storm Surge Prone Areas; (5) V_{half} assumed to be 273 mm/h.

the coefficient in column (1). However, a *t*-test of the difference in the size of coefficients indicates that this difference is imprecisely estimated. Nevertheless, given our assumed damage function and taking the coefficients at face value suggests that a property loss of 50% in a given location reduces economic activity within that area by about 3 units. Given that the average brightness of cells within storm surge areas is higher, it suggests a relative fall in economic activity of about 30%. In contrast, for non-storm surge areas (NSSP) the coefficient in column (4) suggests a 37% smaller impact per percentage of property lost. More precisely, a property loss of 50% at the grid cell level translates into a reduction of 2 units of nightlight intensity in non-storm surge prone areas, or a 20% loss in annual economic activity.[23] Including lags makes little difference to the results and the coefficients on the lag variables continue to be insignificant.

3.2. Functional form

The accuracy of our damage index hinges on three assumptions about its functional form: (i) the cubic relationship between wind speed and damage, (ii) the no-damage threshold, and (iii) the value of wind speed at which the loss ratio is 50%. The assumption of a cubic relationship is a common assumption in the literature and rests on the fact that kinetic energy from a storm dissipates roughly to the cubic power with respect to wind speed and that the energy release scales with the wind pressure acting on a structure.[24] Nevertheless, as part of our robustness checks we attempted to verify this for our data by experimenting with alternative forms of the wind damage relationship. As a benchmark we first re-estimated (7) but using the cubic power of the maximum wind speed above 92 km/h, instead of *f*. The results are shown in the first column of Table 4. As can be seen, the coefficient on this measure is highly statistically significant. In Column (2) we simply use the maximum wind speed in levels (*V*) rather than cubed. We now find no

[23] Note that given the large area of cities relative to the 1 km cells, the average *f* even during very typhoon active years in a city was subsequently relatively low (0.004), while the largest observed value was in Ningde in 2006 (0.182).

[24] See Kantha (2008) and ASCE (2006).

statistically significant impact of wind speed on nightlight intensity. We also experiment using a spline function according to the Saffir–Simpson scale which is commonly used to categorize tropical storm strengths and types of damages. The chosen thresholds correspond to below scale 1 (SS0), scale 1 (SS1), scale 2 (SS2), and scale 3 (SS3).[25] Again, the coefficients, shown in column (3) suggests no significant impact. Assuming a cubic relationship therefore appears to be crucial in capturing damage due to typhoons in China.

In order to determine the wind threshold below which there is unlikely to be any damage one would ideally like to rely on local evidence for China. However, there are no publicly available damage functions for China. We instead refer to FEMA (2010) which has constructed building and loss ratio functions across a large number of building types, distinguished by use, size, roof type and construction, wall type and construction, and the presence of a garage. These loss ratio functions were verified with damage data collected from US Hurricanes Andrew (1992), Eric (1995), and Fran (1996). The results suggest that our 93 km/h cut-off was roughly equivalent to when damages approached zero, no matter what the building or surface type. In column (4) we tested this for China by including, in addition to the cubed maximum wind speed above 93 km/h, a variable of the cubed maximum wind speed, but set to zero above 93 km/h. As can be seen, wind speeds below the cut-off do not significantly affect nightlight intensity.

Finally, an important component of the functional form in Eq. (7) is the assumption of the value for V_{half}, i.e. the assumed value for 50 per cent of damage. Again, FEMA (2010) suggests that the 50 per cent loss ratio is sensitive to the type of building and roughness of the surface in question. More specifically, FEMA (2010) find that it ranges between 190 km/h, and 320 km/h across building types and surface roughness values. With this range of possible 50% damage thresholds in mind, we investigate how sensitive our regression results are to the choice of V_{half}. More specifically, we assume an initial value of 190 km/h and then incrementally increase this by 1 km/h and re-estimated Eq. (7) for each new value. The estimated coefficient and the corresponding 95 per cent confidence bands are shown in Fig. 10. As can be seen, the coefficient on f is significant across the entire range, although the estimates become more imprecise as V_{half} increases. Unsurprisingly, the absolute size of the coefficient also increases with V_{half}. To determine how altering V_{half} affects the estimated impact on nightlights we show in Fig. 11 the implied loss for wind speeds of Saffir–Simpson Scale 3 (178 km/h) using our estimated coefficients. Accordingly, while the reduction in nightlights ascribes to an inverted u-shaped pattern across the range of V_{half}, the differences in nightlight reduction are negligible across the range (about 0.01 units of nightlights, i.e. 0.1% of the average intensity in coastal China). Thus our results are unlikely to be sensitive to our choice of V_{half} of 93 km/h.

3.3. Truncation

A further concern is that our estimates might be affected by the truncation from above of our nightlight variable. For example, if a number of very bright, but upper truncated, cells were damaged, but still retain a value of 63 then this could lead to a downward bias in our estimates. Although only a very small portion of our observations (0.035 per cent or 35,991 cells), ever reached the value of 63, nevertheless, we re-estimate our main specification excluding these observations. The results are shown in the penultimate column of Table 4. As can be seen, this does indeed lower the coefficient on f, however, only marginally (−4.667 instead of −4.694) and we can thus conclude that the upper truncation does not appear to be problematic for our analysis.

3.4. Spatial spillovers

Thus far we have assumed that that tropical winds will only have a direct effect on the cell exposed. Feasibly of course, there may be spillover effects to neighboring cells. In particular, damage in one cell may not only involve building damage but also infrastructure damage thus also affecting the efficiency of transport in neighboring regions. Alternatively, if economic activity is reduced in one area, and neighboring regions, if similar enough, may be able to pick up the slack thus inducing a positive impact on these neighboring regions. To investigate this we calculate the mean value of f of all neighboring cells, up to a maximum of 8 cells, and included this as an additional regression in our main specification. However, as can be seen from the coefficient on f_N in the final column of Table 4 there appears to be no spatial spillover effects of tropical storm damage in Chinese coastal cities.[26] One may want to also note that the coefficient of the own effect has fallen in size. This may not be surprising given the spatial correlation, albeit imperfect, of typhoon wind.

3.5. Disaster preparedness

As a further robustness check we reran the estimations in Table 3 but included two additional controls to capture the possibility that economic activity located in certain cells may be better prepared for typhoon strikes by having invested in, for example, better flood defenses or typhoon resistant buildings. First, we included a variable, *Investment*, which controls for the amount that each city invested in housing in the previous four years. Since the investment data only spans 2001–2008 we first re-estimated our base specification for this shorter time period. Columns (1) and (2) of Table 5 show that our damage index remains a negative and significant determinant of economic activity for cells in both storm surge prone (SSP) and non-storm surge prone (NSSP) areas. When we include our investment variable the coefficient on f continues to be negative and significant for both SSP and NSSP samples. The positive and significant coefficients on investment suggests that investment in housing can reduce the impact of typhoons.

Returning to the full sample 1993–2010, we also experiment with an interaction term of f with its lag in order to see whether recent strikes can increase the sensitivity to a current strike with the results shown in columns (5) and (6). The coefficient on f in NSSP areas is slightly higher but neither the interaction term nor the lag of f term is significant. Finally, in the final three columns, we experiment with the non-exclusion of the zero value cells that were not preceded by a positive intensity value cell. As can be seen, relative to our total sample, this does not alter conclusions concerning the significance of our results and only marginally changes the size of the estimated coefficients compared to Table 3.

3.6. Spatial aggregation bias

Thus far we have only assumed the importance of spatial disaggregation in the analysis. To further investigate this we re-ran our specification at the Chinese county-city level. More specifically, we calculated out the mean nightlight intensity and mean f index for each county-city year and then estimated Eq. (7) for this aggregated sample. The results are presented in Table 6. As can be seen

[25] The maximum wind speed experienced over our sample period was 192 km/h and thus there were no values within the scale categories of 4 and 5.

[26] We thank an anonymous referee for the suggestion to look for a spatial spillover effect.

Table 4
Robustness checks on the functional form of wind speed (dependent variable nightlight intensity).

	(1)	(2)	(3)	(4)	(5)	(6)
V^3	−3.12e−07**			−3.97e−07**		
	(1.06e−07)			(1.28e−07)		
V^3_below92				−6.50e−07		
				(3.48e−07)		
V		−0.003				
		(0.003)				
SS0			−0.009			
			(0.010)			
SS1			0.003			
			(0.009)			
SS2			2.85e−07			
			(3.78e−07)			
SS3			0.009			
			(0.011)			
f					−4.667**	−3.529**
					(1.729)	(1.703)
f_N						−1.240
						(1.542)
Obs.	9,137,607	9,137,607	9,173,598	9,137,607	9,137,607	9,137,607
Cells	563,192	563,192	563,192	563,192	563,192	563,192
R^2	0.300	0.310	0.310	0.310	0.310	0.300
F-test(b)	90.52	100.6	109.0	100.42	102.32	90.52

Notes: (1) Robust standard errors clustered at the city level in parentheses; (2) $^{**}p < 0.01$, $^{*}p < 0.05$; (3) time dummies included but not reported; (4) sample excludes 63 values in column 5; (5) f_N is the average of f in neighboring cells; (6) V is the maximum wind speed experienced; (7) in column 5 V is set to zero below 92 km/h; (8) V_{half} assumed to be 273 km/h in all specifications using f.

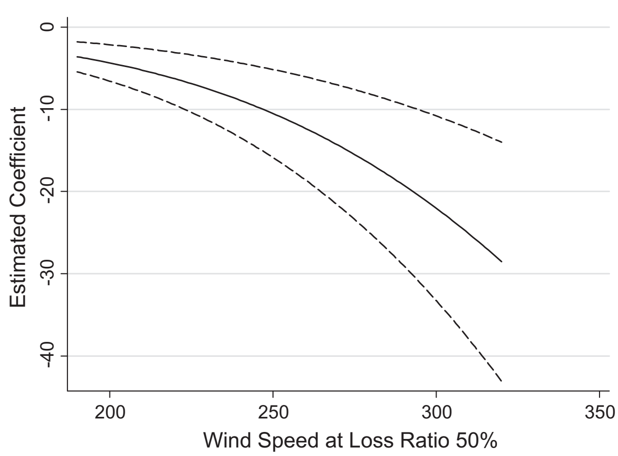

Fig. 10. Estimated coefficient at different V_{half} values (with 95% confidence bands). Notes: (1) Regression coefficient on f assuming different values of V_{half}; (2) dashed lines indicate 95% confidence bands.

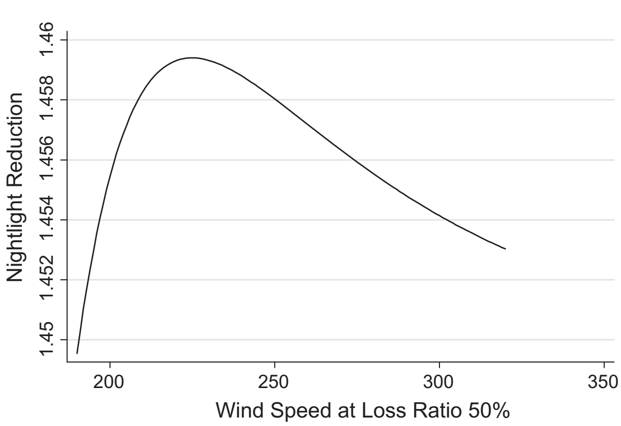

Fig. 11. Nightlight reduction at wind speeds of Saffir–Simpson Scale 3 (178 km/h+) for different V_{half} values. Notes: (1) Implied loss in nightlight intensity using the range of estimated coefficients for different values of V_{half} from Fig. 10 and assuming a maximum wind speed of Saffir–Simpson Scale 3.

in columns (1)–(4) at the county-city level our potential typhoon destruction index no longer has a significant effect on net economic activity for two different V_{half} values (273 km/h and 203 km/h) with and without lags. Part of the reason may be that nightlights are simply not a good proxy for county-city level net economic activity. Hence, in columns (5)–(8) we use city level real GDP/km^2 as an alternative dependent variable, but still find no significant effect for the two wind speeds with and without lags. In the final four columns (9)–(12) we normalize GDP by population rather than by area of a city but still find no significant effect. Our lack of significance at the aggregate level may not be too surprising given that Garrett (2003) has shown that when estimated at a more disaggregated level the estimated effects are statistically significant but heterogeneous, compared to a similar specification at a more aggregate level which is more likely to produce greater standard errors and hence statistical insignificance.[27] More generally this exercise demonstrates the importance of using as spatially disaggregated data as possible when seeking to estimate the economic effects of natural disasters since in more aggregate data the impact of a natural disaster may simply be ‘aggregated’ out.[28]

4. Economic significance

To obtain a better understanding of the economic importance of typhoons ideally one would like derive actual monetary values for the loss in (net) economic activity due to typhoons. In order to translate our estimates into rough monetary values we first calculate the average nightlight intensity for our coastal cities and merge this data with the available data on GDP per square kilometers from the county-city database. We then estimate the following:

$$GDP/AREA_{it} = \alpha + \beta_{NL} Nightlight_{it} + \pi_t + \mu_i + \varepsilon_{it} \quad (8)$$

[27] We are controlling for individual heterogeneous effects through the use of cell level fixed effects in our nightlight cell level regressions.

[28] See Cole et al. (2013) for a disaggregated approach to estimating the effects of the 1995 Kobe earthquake on manufacturing plant performance.

Table 5
Further robustness checks on disaster preparedness (dependent variable nightlights).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
f	−7.564**	−2.890*	−7.415**	−3.615**	−7.042*	−4.191*	−5.290**	−3.991**
	(0.868)	(1.189)	(0.961)	(1.073)	(3.454)	(1.928)	(2.748)	(1.583)
$f(t-1)$					−1.828	−3.216	−0.849	−2.670
					(3.114)	(2.523)	(2.61)	(1.931)
$f*f(t-1)$					26.952	25.453		
					(27.684)	(20.615)		
Investment			0.00371**	0.00344**				
			(0.000962)	(0.00101)				
Sample	SSP	NSSP	SSP	NSSP	SSP	NSSP	SSP	NSSP
Period	2001–2008	2001–2008	2001–2008	2001–2008	1993–2010	1993–2010	1993–2010	1993–2010
Obs.	1,092,255	3,079,286	1,092,255	3,079,286	2,425,384	9,173,598	2,523,816	7,613,640
Cells	137,729	398,843	137,729	398,843	140,212	563,192	140,212	563,192
R^2	0.283	0.179	0.306	0.191	0.400	0.283	0.391	0.2775
F-test(b)	120.5	79.83	93.16	87.67	85.61	118.37	65.83	121.56

Notes: (1) Robust standard errors clustered at the city level in parentheses; (2) $^{**}p < 0.01$, $^{*}p < 0.05$; (3) time dummies included but not reported; (4) SSP: Storm Surge Prone Areas; NSSP: Non SSP; (4) investment is city level investment in housing.

Table 6
Further robustness checks on spatial aggregation bias (dependent variable city level nightlight intensity and city level GDP).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
f	−9.295	−9.324	−2.910	−2.65	−0.254	−0.235	−0.0724	−0.067	−0.569	−0.576	−0.148	−0.151
	(4.875)	(4.923)	(1.596)	(1.68)	(0.148)	(0.141)	(0.0438)	(0.042)	(1.211)	(1.191)	(0.399)	(0.392)
$f(t-1)$		−4.758		−1.321		−0.336		−0.106		0.423		0.169
		(4.842)		(1.505)		(0.207)		(0.0670)		(1.861)		(0.606)
Dep. Var.	NL	NL	NL	NL	GDP	GDP	GDP	GDP	GDP	GDP	GDP	GDP
V_{half} (km/h)	273	273	203	203	273	273	203	203	273	273	203	203
Nr. Cities	59	59	59	59	58	58	58	58	58	58	58	58
Obs.	1062	1062	1062	1062	865	865	865	865	865	865	865	865
R-squared	0.665	0.665	0.665	0.665	0.099	0.102	0.099	0.102	0.102	0.074	0.075	0.075

Notes: (1) Robust standard errors clustered at the city level in parentheses; (2) $^{**}p < 0.01$, $^{*}p < 0.05$; (3) time dummies included but not reported; (4) NL: average nightlights; (5) GDP: real GDP per square km in columns 5 to 8, and GDP per capita in columns 9 to 12.

where μ are coastal county-city fixed effects, and π are time dummies. It is not our intention that Eq. (8) should be viewed as a causal regression, but rather a simple method to extract the correlation of nightlights and GDP data at the county-city level while controlling for county-city and time specific factors. The resultant estimated coefficient on *Nightlight* is found to be 0.036 which is significant at the 1% level. Hence, the correlation suggested visually in Fig. 4 is confirmed statistically. Taken more precisely, our results indicate that each unit of nightlight per squared kilometer corresponds to 0.036 of income per square kilometer. Using this coefficient and the coefficient from the first column of Table 3 implies that a typhoon that causes 20% destruction of property reduces GDP/AREA by 0.01 (i.e. the product of f, β_{NL}, and β_f). Given that GDP/AREA is measured in per km^2 and has a mean of 0.1401, our results suggest that a typhoon that causes 20% destruction reduces mean income by about 14% in that year. For cells within a storm surge prone area the equivalent value would be about 18.6%.

We can also use the relationship between GDP/km^2 and nightlights to calculate the total losses in net economic activity over our sample period. More specifically, we estimate the total losses in any year to be equal to:

$$Loss_t = \sum_{i=1}^{N} nightlight_{t-1} f_t \beta_f \beta_{NL} \tag{9}$$

for all cells $i = 1,\ldots,N$ in our sample. Total losses in any time t will thus depend not only on the distribution of local maximum wind speeds through f, but also on the local and total amount of exposure in terms of nightlight intensity. For example, even if the relative distribution of maximum wind speeds for a storm is the same within two cities, the smaller city may still experience a larger total reduction in economic activity due to incidental greater exposure in the more wind intensive cells.

Applying Eq. (9), and allowing for different estimated β_{NL} for cells in SSA and NSSAs, we find that over our sample period, losses totaled $US 28.34 billion which translates into an average annual loss of $US 1.44 billion. Note that these losses were the result of the impact from a total of 78 damaging storms over our period. Within our period of analysis the top three storms in terms of damage (Khanun, Saomai, and Rananim) caused approximately $US 7 billion of total net losses, while of the remaining 75, only four produced more than a $US 1 billion reduction in net economic activity. Thus, despite the publicity that results from large typhoons the majority of the impact is still a result of the large number of storms that strike the Chinese coast.

In Fig. 12 we graph the loss estimates to show that in general, annual net losses due to damaging typhoons appear to have increased over time. As has been found for hurricane damages in the US by Pielke et al. (2008), it is however likely that some of the rising costs of tropical storms over the years in China have been due to greater concentration of economic activity on the coast (as can be seen from a comparison of Figs. 2 and 3). To roughly assess how much of a role such increasing vulnerability might have played we took Eq. (8) and set $nightlights_t = nightlights_{1992}$, i.e. we set the intensity of nightlights as if they had remained at their 1992 values. This means assuming constant zero nightlight intensity for 47% of our cells that actually became positive at some point after 1992. As can be seen from the blue bars in Fig. 12, increased vulnerability played a large role as an explanation of annual net losses. Overall, losses would have been about $US 10.3 billion, i.e. about 36% of the previous estimate. More generally, this simple exercise suggests that net losses in economic activity would not have been too different in the latter compared to the former years if the coastal concentration of economic activity had not changed much since 1992.

One can also use also our estimates to identify differences across county-cities. When we do so we find that since 1992 40 out of a total of 60 administrative areas have suffered net losses in economic activity due to typhoon damage. On average, total losses were $US 0.48 billion per county-city. In terms of losses by individual cities, Taizhou, Wenzhou, and Zhanjiang rank the highest, with total net losses of $US 4.5 billion, $US 3.54 billion and $US 2.41 billion, respectively.

5. Risk analysis

One drawback with our historical analysis is that when considering disaster prevention and/or mitigation measures policy makers are likely to be interested in expected future losses rather than what has happened in the past. This would involve not only knowing the impact of typhoons but also being able to determine the probabilities of future typhoons and their likely strength. In this regard one might be tempted to simply use all available historical data to derive a probability distribution. However, even though typhoon track data in the NWP is available as far back as 1950, the actual number of historical extreme events is not sufficient enough to accurately determine a probability distribution of the return period and potential damages of typhoons in the area.[29]

Early studies countered the extreme events problem by fitting the data to standard distributions, such as the lognormal or Weibull, and then drawing randomly from these to generate a large enough set to conduct a risk assessment. However, this method is sensitive to the tail of the assumed distribution for which the historical data can at best only normally provide a few data points. This weakness was partially addressed by using empirical global distributions of relative intensity in conjunction with climatological values of potential intensity to derive local intensity distributions (Murnane et al., 2000). In contrast, Jagger and Elsner (2006) tackled the tail problem by fitting the data to functions that conform to expectations based on generalized extreme-value theory, but again this can be sensitive to the assumed attributes of the distribution.

In this paper we employ another, recent and increasingly popular, statistically deterministic approach following Emanuel et al. (2008) that generates a large set of synthetic tropical storm events. The Emanuel et al. (2008) approach takes recent historical climate data within a coupled ocean–atmosphere tropical storm model to generate hypothetical tropical storms. More specifically, tracks are created on the basis of meteorological global information on, amongst other things, temperature, humidity, wind, and sea surface temperature and temperature. First proto-storms are placed randomly in space and time and are assumed to move with the large-scale winds of a given climate state. Their intensity is then determined using a coupled ocean–atmosphere tropical storm model. Eventually a small set of these storms develop into full scale typhoon strength tropical storms. Using this methodology we utilize 3000 synthetically generated typhoon strength storms traversing the Northwestern Pacific Basin region on the basis of yearly weather data (wind, sea surface temperature, air temperature, and humidity) observed over the 1980–2010 period.[30] Each storm is assigned to one of the 30 years climatology and each year has a given expected frequency of storms. The model provides, amongst other things, the location of the eye, the forward velocity, the central

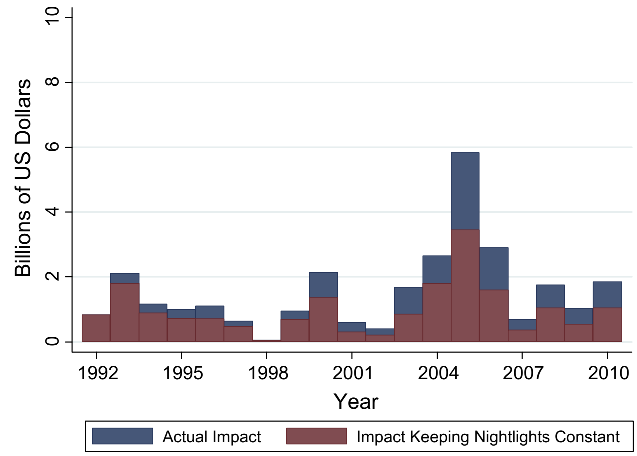

Fig. 12. Reduction in economic activity (billions of $US). Notes: (1) *Actual Impact:* Sum of the observed value of f at time t times the estimated coefficient of f times the value of nightlight at $t-1$ over all coastal city cells at time t. (2) *Impact Keeping Nightlight Constant:* Sum of the observed value of f times the estimated coefficient of f at time t times the value of nightlight at 1992 over all coastal city cells for year t.

pressure, and the radius of maximum winds for every two hours of a storm's lifetime. Maximum wind speed among the synthetic storms varies between 119 and 312 km/h, with a mean of 139 km/h. Thus we have at hand a large set of possible typhoon strength tropical storms in the Northwestern Pacific Basin close enough to China to potentially cause damages that might have formed if the weather was similar to the 1980–2010 period.[31]

In order to generate a probability distribution of storm events we first assumed that the probability of each year's weather over the 1980–2010 period was equally likely. To construct a specific year of events we then randomly picked a year and, using a Poisson distribution, randomly drew from each year, according to the expected frequency of events corresponding to that year, a set of storms. This was done a total of 100,000 times, generating a total of 374,085 storms, for each of which we can calculate an annual probability. Note that in these 100,000 hypothetical storm-years, 23 per cent produced no damage. This suggests that if we observe similar weather to that experienced in the last year 30 years there would on average be a typhoon damage free year every four years and demonstrates the unusually large number of damaging storms that took place over our sample period, where no year was damage free. This difference emphasizes the difficultly in making predictions concerning future tropical cyclone activity with historical data, in that a few extreme, but very unlikely events may be driving past estimates. For those years with positive damage, the damage estimate was on average $US 0.75 billion, with a large standard deviation of $US 8.6 billion, a median of $US 0.05 billion, and a maximum of $US 303 billion.[32]

Hence, we now use Eqs. (1)–(6) to calculate the maximum value of f for each cell for each synthetic storm and, given its occurrence in our simulated data set, attached a probability of occurrence to each event for each cell.[33] Using the probability of occurrence Fig. 13 shows the likely return period of a damaging typhoon across the entire set of cells on the Chinese coast. As can be seen, return periods differ widely across coastal county-cities. In particular,

[29] Moreover, even if the number were sufficient using the whole historical track data set, the weather driving tropical storm formation is unlikely to have been stable over the last 60 years.

[30] We are grateful to Kerry Emanuel for kindly generating these tracks for us.

[31] Emanuel et al. (2008) compared their models to historical data and found that their model was a good predictor of historical patterns.

[32] Note that this difference is only marginally driven by the fact that the synthetic storms are generated from weather outside of our sample period. When we exclude those years not corresponding to our data (1980–1991) the proportion of zero damaging years decreases only slightly to 19 per cent.

[33] The only difference is that the synthetic track generator provides a value of R_{max} which means it no longer has to be estimated.

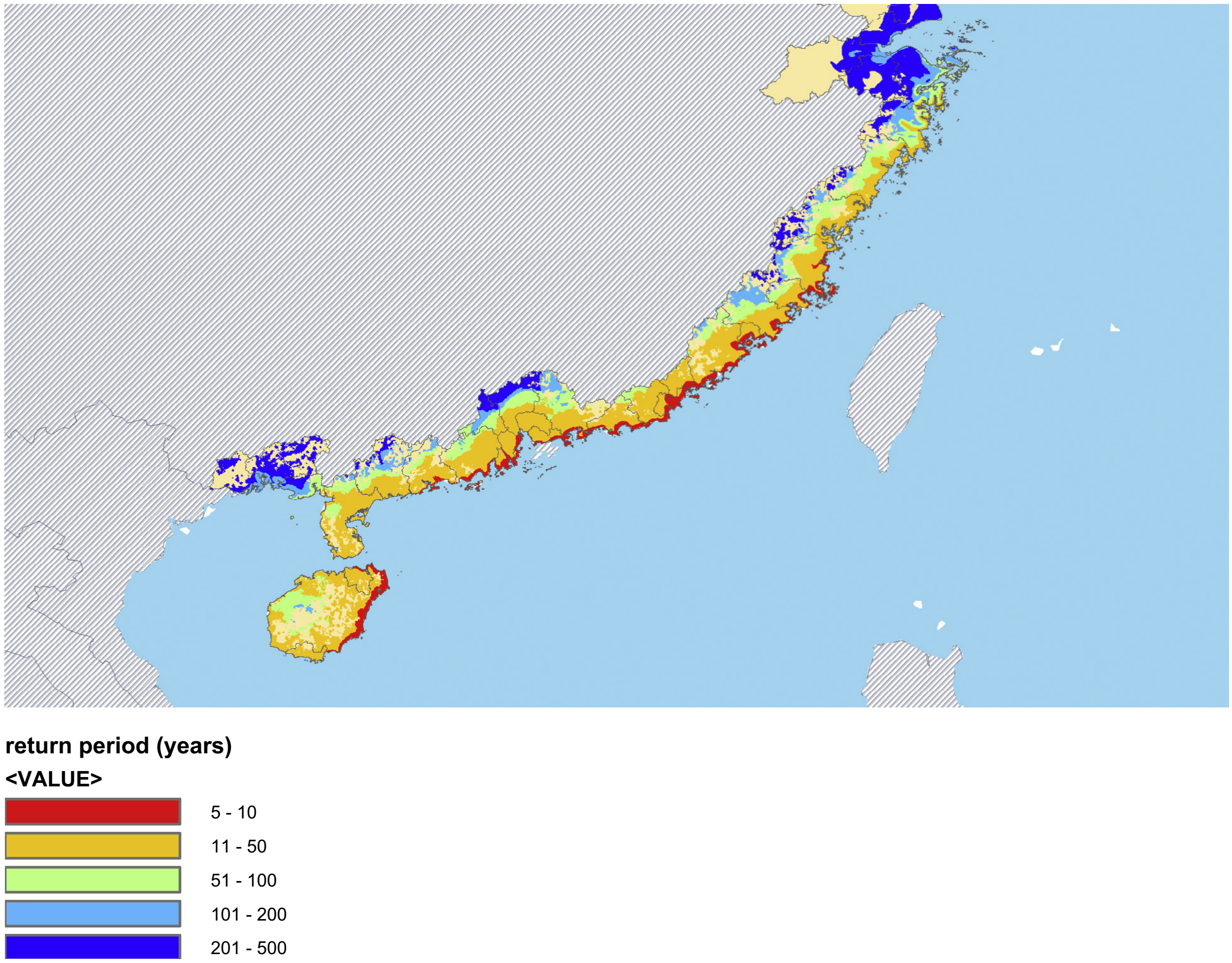

Fig. 13. Return period of damaging typhoon ($f > 0$). Notes: (1) Return period of a damaging typhoon estimated using simulated tracks.

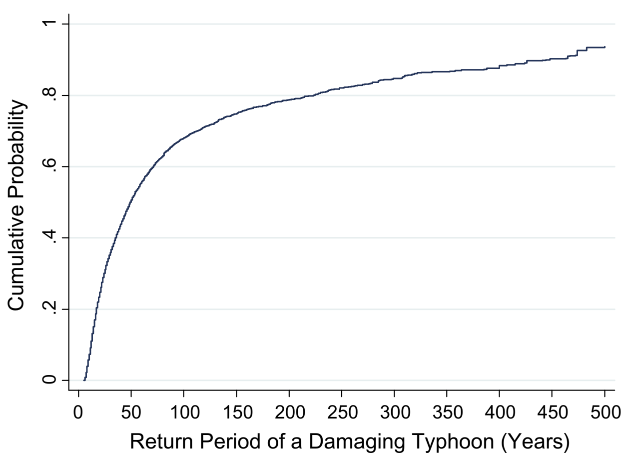

Fig. 14. Cumulative probability of the return period of damaging typhoons. Notes: (1) Cumulative probability of the return period of a damaging typhoon estimated using simulated tracks.

return periods of between 5 to 10 years should be expected for regions very close to the coast and much of the southern coast of China. In Fig. 14 we also show the cumulative distribution of return periods across cells up to 500 years. Accordingly, about 20 per cent of cells are likely to experience a damaging typhoon within the next 20 years. If one considers the next 100 years this figure rises to about 70 per cent, whereas almost all cells are likely to be hit by a typhoon within the next 500 years.

Our estimates can of course also be translated into monetary values using our estimated coefficients. We find that about 44 per cent of all cells have a non-zero expected value of annual damages. For 90% of these areas, the expected loss, i.e., the probability of occurrence times the damage, is less than $US 0.16 million, but with considerable variance across cells, where the highest value is over $US 1.7 million. In more aggregated terms our simulated tracks suggest that net annual expected reduction in economic activity is about $US 0.54 billion. It is interesting to observe that this value is lower than that calculated for our historical sample period and serves to emphasize the importance of having a large number events to form expectations about the economic impact of typhoons. Otherwise pure chance could mean that the expected value calculated from historical data could be dominated by a few damaging, but very unlikely, events.

When we consider the county-city level we also find that expected annual losses differ widely, with those in Hainan ($US 116 million), Zhanjiang ($US 63 million), Maoming ($US 38 million), and Quinzhou ($US 28 million) being the highest, while at the lower end cities like Jiaxing, Suzhou, and Nantong are all likely to experience annual losses of less than $US 80,000. In this regard it is reassuring that our analysis supports the Guy Carpenter Instrat (2006) conclusion that "*. . .different regions are exposed to different degrees of risk, with the highest risk located along the southern and eastern coast*" (page 2).

6. Conclusions

In this paper we combined satellite imagery data with historical and simulated typhoon tracks to provide highly spatially disaggregated estimates of the impact of typhoons on local economic activity along the coast of China. Our econometric results show that typhoons had a significant negative, but short-lived effect. In translating our coefficient estimates into monetary values we found that total losses in terms of GDP amounted to about $US 1.44 billion a year on average, where this increase in costs over time is to a considerable extent due the rapid economic Chinese coastal development since 1992. Using simulated typhoon tracks to conduct a risk analysis we show that some local areas should expect the return of a damaging storm every five years, whereas others are likely to be hit no more than every 500 years. In more aggregate terms, expected annual losses are likely to be around $US 0.54 billion for costal China, although losses differ widely by city.

Given our finding of significant net reductions in economic activity as a consequence to typhoon strikes, as well as likely further large reductions in the future, the important issue from a policy perspective is how to minimize these potential future losses. One drawback of our analysis is of course that we are not able to disentangle the role of post disaster management in dampening the effects, particularly since this is viewed as having been fairly successful in China. This should be an avenue for future research. Regardless, the concern, however, as pointed out by Burton (2004), is that there still exists an ''adaption deficit'', where society responds only slowly to increases in disaster exposure. In this regard, China night want to take into account the rising incomes and increasing population pressure in coastal regions that have increased vulnerability substantially over the last decades. This may mean looking carefully at land use and urban planning and trying to build infrastructural and societal resilience to typhoons.

Acknowledgments

We would like to thank John Mutter, Ralph Martin, Pierre-Philippe Combes, Matthew Cole, two anonymous referees and seminar participants at the Grantham Institute (LSE) and the Earth Institute (Columbia University) for useful comments. We would also like to thank Kerry Emanuel for help with the data. The usual disclaimer applies.

References

Annan, K.A., 1999. An Increasing Vulnerability to Natural Disasters, New York Times Opinion, September. <http://www.nytimes.com/1999/09/10/opinion/10iht-edannan.2.t.html>.

ASCE, 2006. Minimum Design Loads for Buildings and Other Structures, ASCE/SEI 7-05. American Society of Civil Engineers, Reston, VA.

Asia Disaster Reduction Centre (ADRC), 2013. <http://www.adrc.asia/nationinformation.php?NationCode=156&Lang=en&NationNum=22> (accessed 15.08.13).

Benson, C., Clay, E., Bank, W., 2000. Developing countries and the economic impacts of natural disasters. In: Kreimer, Alcira, Arnold, Margaret (Eds.), Managing Disaster Risk in Emerging Economies, Disaster Risk Management Series 2. World Bank, Washington, DC, pp. 11–21.

Bertinelli, L., Strobl, E., 2013. Quantifying the local economic growth impact of hurricane strikes: an analysis from outer space for the Caribbean. Journal of Applied Meteorology and Climatology 52, 1688–1697.

Boose, E., Serrano, M., Foster, D., 2004. Landscape and regional impacts of hurricanes in Puerto Rico. Ecological Monograph 74, 335–352.

Bouwer, L., 2011. Have disaster losses increased due to anthropogenic climate change. Bulletin of the American Meteorological Society, 39–46.

Brecht, H., Dasgupta, S., Laplante, B., Murray, S., Wheeler, D., 2012. Seal-level rise and storm surges: high stakes for a small number of developing countries. The Journal of Environment and Development 21, 120–138.

Burton, I., 2004. Climate change and the adaptation deficit. Environment Canada Occasional Paper 1, 7 pp.

Cao, S., Wang, J., 2013. Statistical summary and case studies of strong wind damage in China. Journal of Disaster Research 8 (6), 1096–1102.

Cavallo, E., Noy, I., 2011. Natural disasters and the economy – a survey. International Review of Environmental and Resource Economics 5, 63–102.

Chen, X., Nordhaus, W.D., 2011. Using luminosity data as a proxy for economic statistics. Proceedings of the National Academy of Sciences of the United States of America 108, 8589–8594.

Chen, P., Yang, Y., Lei, X., Qian, Y., 2009. Cause analysis and preliminary hazard estimate of typhoon disaster in China. Journal of Natural Disasters 18 (1), 64–73.

Cole, M.A., Elliott, R.J.R., Okubo, T., Strobl, E., 2013. Natural Disasters and Plant Survival: The Impact of the Kobe Earthquake, RIETI Discussion Paper 2013/04.

Elsner, J., Kossin, J., Jagger, T., 2008. The increasing intensity of the strongest tropical cyclones. Nature 455, 92–95.

Elvidge, C.D., Baugh, K.E., Kihn, E.A., Kroehl, H.W., Davis, E.R., Davis, C.W., 1997. Relation between satellites observed visible – near infrared emissions, population, economic activity and electric power consumption. International Journal of Remote Sensing 18 (6), 1373–1379.

Emanuel, K.F., 2011. Global warming effects on US hurricane damage. Weather, Climate, and Society 3, 261–268.

Emanuel, K.F., Sundarajan, R., Williams, J., 2008. Hurricanes and global warming: results from downscaling IPCC AR4 simulations. Bulletin of American Meteorological Society 89, 347–367.

Federal Emergency Management Authority [FEMA], 2010. HAZUS-MH MR5 Technical Manual. Washington, DC.

Felbermayr, G., Gröschl, J., 2013. Naturally Negative: The Growth Effects of Natural Disasters, CESifo Working Paper No. 4439.

Forbes, C., Luettich, R., Mattocks, C., 2011. Storm surge simulations of Hurricane Ike (2008): its impact Lousiana and Texas. In: Spaulding, M. (Ed.), Estuarine and Coastal Modeling. ASCE.

Freeman, P.K., Keen, M., Mani, M., 2003. Dealing with Increased Risk of Natural Disasters: Challenges and Options, IMF Working Paper 03/197. Washington DC, United States.

Fritz, H., Blount, C., Sokoloski, R., Singleton, J., Fuggle, A., MacAdoo, B., Moore, A., Grass, C., Tate, B., 2007. Hurricane Katrina storm surge distribution and field observations on the Mississippi Barrier Islands. Estuarine, Coastal and Shelf Science 1, 1–9.

Gao, J., Zhu, X., Yu, Y., Jin, B., 1999. Study of the impact of typhoon disaster on coastal region of China. Journal of Catastrophology 14 (2), 73–77.

Garrett, T., 2003. Aggregated versus disaggregated data in regression analysis: implications for inference. Economics Letters 81, 61–65.

Guy Carpenter Instrat, 2006. Typhoon Saomai: Impact and Historical Comparison, Report.

Harari, M., La Ferrara, E., 2013. Conflict, Climate and Cells: A Disaggregated Analysis, CEPR Discussion Papers 9277, CEPR Discussion Papers.

Henderson, V., Storeygard, A., Weil, D.N., 2011. A bright idea for measuring economic growth. American Economic Review 101 (3), 194–199.

Hodler, R., Raschky, P., 2014. Regional favoritism. Quarterly Journal of Economics 129, 995–1033.

Holland, G., 1980. An analytic model of the wind and pressure profiles in hurricanes. Monthly Weather Review 106, 1212–1218.

Holland, G., 2008. A revised hurricane pressure-wind model. Monthly Weather Review, 3432–3445.

Im, K.S., Pesaran, M.H., Shin, Y., 2003. Testing for unit roots in heterogeneous panels. Journal of Econometrics 115, 53–74.

Jagger, T.H., Elsner, J.B., 2006. Climatology models for extreme hurricane winds near the United States. Journal of Climate 19, 3220–3236.

Kahn, M., 2005. The death toll from natural disasters: the role of income. Geography, and Institutions, Review of Economics and Statistics 87 (2), 271–284.

Kantha, L., 2008. Tropical cyclone destructive potential by integrated energy. Bulletin of the American Meteorological Society 89, 219–221.

Landsea, C.W., Harper, B.A., Hoarau, K., Knaff, J.A., 2006. Can we detect trends in extreme tropical cyclones. Science 313 (5785), 452–454.

Li, T., Kwon, M.H., Zhao, M., Kug, J.S., Yu, W., 2010. Global warming shifts pacific tropical cyclone location. Geophysical Research Letters 37, L21804.

Liu, K.B., Shen, C., Louie, K.S., 2001. A 1,000-year history of typhoon landfalls in Guangdong, Southern China, reconstructed from Chinese historical documentary records. Annals of the Association of American Geographers 91 (3), 453–464.

Louie, K.S., Liu, K.B., 2003. Earliest historical records of typhoons in China. Journal of Historical Geography 29, 299–316.

McGranahan, G., Balk, D., Anderson, B., 2007. The rising tide: assessing the risks of climate change and human settlements in low elevation coastal zones. Environment and Urbanization 19, 17–37.

McKinsey Global Institute, 2009. Preparing for China's Urban Billion. <http://www.mckinsey.com/insights/mgi/research/urbanization/preparing_for_urban_billion_in_china> (accessed November 2012).

Mendelsohn, R., Emanuel, K., Chonabayashi, S., Bakkensen, L., 2012. The impact of climate change on global tropical cyclone damage. Nature Climate Change 2, 205–8209.

Meng, F., Kang, J., Li, W., Wu, T., Wang, T., An, Y., 2007. Analysis and evaluation of typhoon disasters in Shanghai in past 50 years. Journal of Catastrophology 22 (4), 71–76.

Michalopoulos, S., Papaioannou, E., 2014. National institutions and subnational development in Africa. Quarterly Journal of Economics 129 (1), 151–213.

Murnane, R.J. et al., 2000. Model estimates hurricane wind speed probabilities. Eos, Transactions American Geophysical Union 81, 433–438.

National Bureau of Statistics, 2013.

Neumayer, E., Barthel, F., 2011. Normalizing economic loss from natural disasters: a global analysis. Global Environmental Change 21, 13–24.
Nicholls, R.J., 2006. Storm surge in coastal areas, natural disasters hot spots: case studies. In: Arnold, M. et al. (Eds.), World Bank Disaster Risk Management, vol. 6. World Bank, pp. 79–108.
Paulsen, B., Schroeder, J., 2005. An examination of tropical and extratropical gust factors and the associated wind speed histograms. Journal of Applied Meteorology 44, 270–280.
Pielke Jr., R.A., Gratz, J., Landsea, C.W., Collins, D., Saunders, M.A., Musulin, R., 2008. Normalized hurricane damages in the United States: 1900–2005. Natural Hazards Review 9 (1), 29–42.
Polyzos, S., Tsiotas, D., 2012. The evolution and spatial dynamics of coastal cities in Greece. In: Polyzos, S. (Ed.), Urban Development. In Tech. ISBN: 978-953-51-0442-1. <http://www.intechopen.com/books/urban-development/the-evolution-and-spatial-dynamics-of-coastal-cities-in-greece>.
Strobl, E., 2008. The Economic Growth Impact of Hurricanes: Evidence from U.S. Coastal Counties, IZA Discussion Paper, 3619.
Strobl, E., 2011. The economic growth impact of hurricanes: evidence from U.S. coastal counties. Review of Economics and Statistics 93, 575–589.
Strobl, E., 2012. The economic growth impact of natural disasters in developing countries: evidence from hurricane strikes in Central American and Caribbean regions. Journal of Development Economics 97, 130–141.
Swiss Reinsurance Company, 2005. World Insurance in 2004: Growing Premiums and Stronger Balance Sheets, Sigmas No. 2/2005.
Tse, C.W., Wei, J., Wang, Y., 2013. Social Capital and Disaster Recovery: Evidence from Sichuan Earthquake in 2008, Centre for Global Development, Working Paper 344.
UNISDR, 2009. Global Assessment Report on Disaster Risk Reduction, UNISDR, pp. 207.
Vickery, P., Wadhera, D., 2008. Statistical models of holland pressure profile parameter and radius to maximum winds of hurricanes from flight-level pressure and H*Wind data. Journal of Applied Meteorology and Climatology 47, 2497–2517.
Vickery, P., Masters, F., Powell, M., Wadhera, D., 2009. Hurricane hazard modeling: the past, present, and future. Journal of Wind Engineering and Industrial Aerodynamics 97, 392–405.
Vu, T.B., Hammes, D., 2010. Dustbowls and high water, the economic impact of natural disasters in China. Asia Pacific Journal of Social Science 1, 122–132.
Vu, T.B., Noy, I., 2013. Regional Effects of Natural Disasters in China, Working Paper Series, Victoria University of Wellington, School of Economics and Finance 2812, Victoria University of Wellington, School of Economics and Finance.
Wang, G., Su, J., Ding, Y., Chen, D., 2007. Tropical cyclone genesis over the South China Sea. Journal of Marine Systems 68, 318–326.
Ward, R.E.T., Herweijer, C., Patmore, N., Muir-Wood, R., 2008. The role of insurers in promoting adaptation to the impacts of climate change. The Geneva Papers on Risk and Insurance – Issues and Practice 33, 133–139.
Webster, P.J., Holland, G.J., Curry, J.A., Chang, H.R., 2005. Changes in tropical cyclone number, duration, and intensity. Science 309 (5742), 1844–1846.
Wu, D., 2004. Report for the Development and Reform of China's Insurance Industry: 1979–2003. China Economic Publishing House, Beijing (in Chinese).
Xiao, Yu-Feng, Xiao, Y-Quing, Duan, Zhong-Dong, 2009. The Typhoon Wind Hazard Analysis in Hong Kong of China with the New Formula for Holland B Parameter and the CE Wind Field Model, The Seventh Asia-Pacific Conference on Wind Engineering, Nov. 8–12, Taipei, Taiwan.
Zhang, C., Zhu, N., 2005. Determinants of the Development of Insurance in China under Globalization. <www.cerdi.org/colloque/CHINE2005/papier/Zhang_31.pdf>.
Zhang, L., Liu, M., Quan, R., Lu, M., Wang, J., Xu, S., 2009a. Characteristics and disaster valuation of the tropical cyclones in southeast coastal areas of China. Journal of East China Normal University (Natural Science) 02, 41–49.
Zhang, Q., Wu, L., Liu, Q., 2009b. Tropical cyclone damages in China: 1983–2006. Bulletin of the American Meteorological Society 90, 489–495.